Power Device Reference Board

3.6kW Interleaved Boost PFC Reference Design

1. Introduction

This application note provides an overview of ROHM 3.6 kW Interleaved Boost PFC Reference Design **(REF67010-IPFC_3.6kW-EVK-A04)**. The board has a wide input AC voltage between 180 Vac to 265 Vac and provides a 390 V output voltage, PFC offers: reduced losses in the device during switching, better voltage regulation, increased capacity to serve power requirements, Lower energy for each phase, high efficiency, improved PF and THD.



Figure 1. 3.6 kW Interleaved Boost PFC (IPFC) reference board (Top view).

As shown in **Figure.1**, **REF67010-IPFC_3.6kW-EVK-A04** is the 3.6 kW Interleaved Boost PFC (**IPFC**) reference board. The reference board showcases **ROHM's** strong power devices i.e., high speed switching device Field Trench IGBT and SiC Schottky Barrier Diodes (**SBD**) in the boost stage. The Auxiliary Power Supply (**AuxPS**) uses PWM type DC/DC converter IC with 800 V Integrated MOSFET and Super-Fast Recovery Diode (**FRD**).

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Block Diagram

The functional **block diagram** of the **Reference Board** (**REF67010-IPFC_3.6kW-EVK-A04**) is shown in **Figure 2**. The key design blocks are the EMI filter and rectification blocks converting AC into DC Voltage. The rectified voltage is then delivered to Interleaved Boost PFC (IPFC) block. The IPFC section uses two low side Gate driver Integrated Circuit (**IC**). The PWM signal in the Gate driver IC switches the two MOSFETs 180° out of phase. The output capacitors C_{bulk} absorbs the AC component of the current, charge the capacitors and delivers power to the load.

The Auxiliary Power Supply (**AuxPS**) block is designed with **ROHM**'s PWM type DC/DC converter IC **(BM2P063HK-LBZ**) that contains inbuilt 800 V MOSFET and the gate driver. The output voltages of the power supply are 15 V (for Low side gate driver IC's), 12 V (for Relay inrush current limit), and 3.3 V (for Digital signal control).

The protection features implemented include soft-start, input over voltage and under voltage, output over current, output over voltage and leg current circuit protection

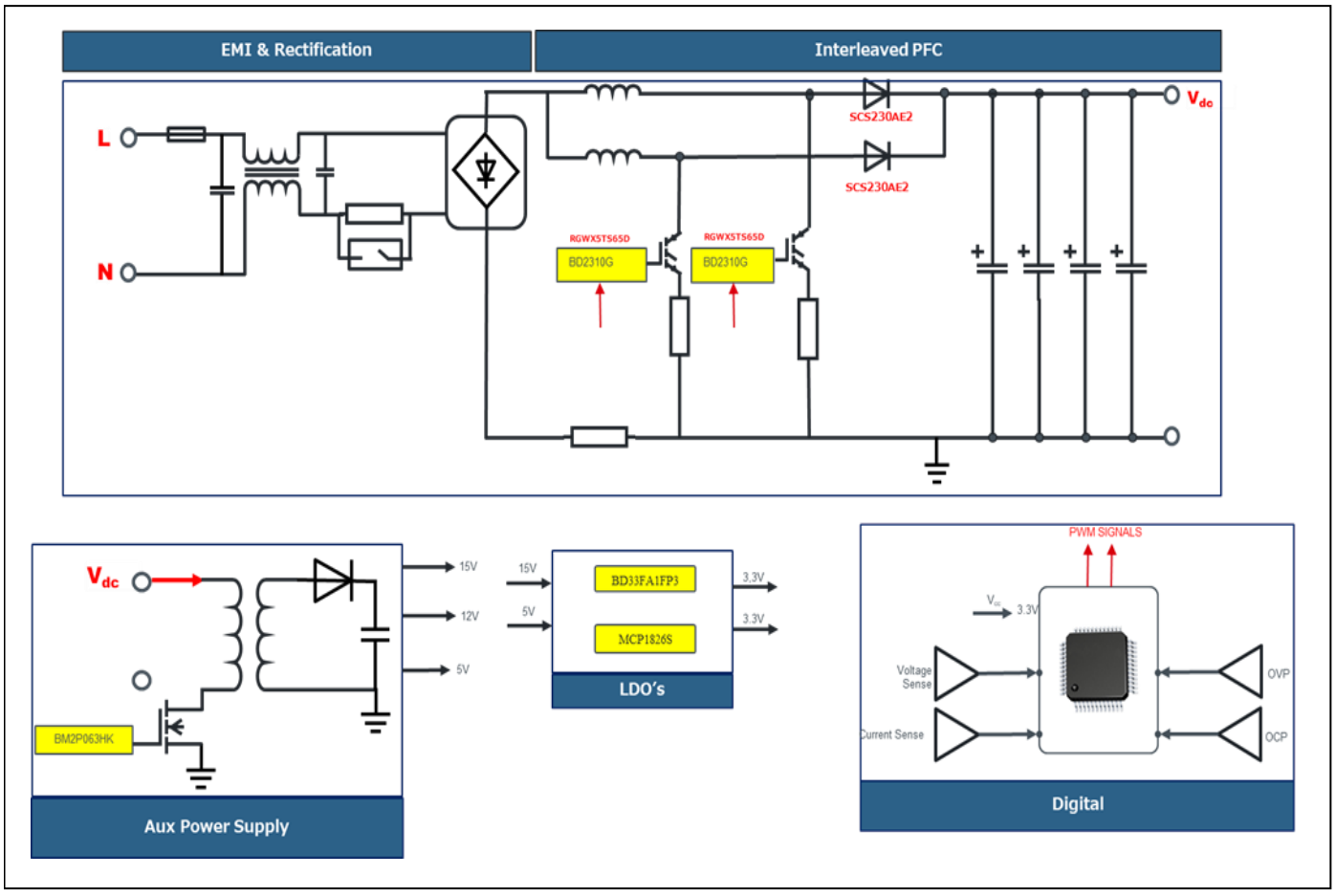


Figure 2. Block diagram of 3.6 kW Interleaved Boost PFC (IPFC).

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Main Components

Table 1: **ROHM** Components used in 3.6 kW Interleaved Boost PFC (IPFC).

Section	Part Number	Function	Function
IPFC Boost	RGWX5TS65D	IPFC Switch	High-Speed Fast Switching Type, 650V 75A, FRD Built-in, TO-247N, Field Stop Trench IGBT
IPFC Boost	SCS230AE2	Boost Diode	650V, 30A, 3pin THD, Silicon-carbide (SiC) SBD
IPFC Driver	BD2310G	PFC Gate Driver	1ch 4A High Speed Low-side Gate Driver
IPFC Driver	BD33FA1FP3	LDO	100mA 3.3V, Fixed Output, High-Accuracy LDO Regulator
Auxiliary Power Supply	BM2P063HK	DC/DC converter IC	Integrated 800V Switching MOSFET, 65kHz, PWM Type DC/DC Converter IC
IPFC Boost	RB068LAM-60	Protection Diode	Super Low I_R, 60V, 2A, Schottky Barrier Diode
Auxiliary Power Supply	RF302LAM2S	Flyback secondary diode	Super-Fast Recovery Diode with low V_f and low switching loss

Table 2: 3.6 kW Interleaved Boost PFC (IPFC) specifications.

Parameter	Units	Min	Typ	Max
AC Input Voltage	V_{AC}	180	230	300
AC Input Frequency	Hz	47	50	60
Output Power	W	3500	3600	3780
DC Bus Voltage	V_{DC}	350	390	410
Full Load Efficiency	%	95	96	97
PF		0.96	0.98	0.99

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2. Interleaving Boost PFC

Boost PFC is the most popular topology used for power factor correction due to the presence of an input inductor that enables the input continuous current operation. The current mode control technique can be employed to track the current and also to bring it in line with the input voltage.

To further improve the power density and the magnetics volume, interleaving can be employed. This is done by having 'n' branches of similar ratings, so that the current through each branch is reduced by '1/n' times. This leads to reduced volume of magnetics, reduced RMS current in the boost capacitor. The standout advantage of the interleaved PFC is the inductor ripple current cancellation. Since the branches are operating 180° out of phase, the input current is the sum of the inductor currents. The best inductor current ripple current cancellation occurs at 50% duty cycle.

The ratio of the input current ripple to the inductor current ripple (K(D)) shows the effect of duty cycle on the input current ripple.

$$K(D) = \frac{\Delta I_{in}}{\Delta I_{L1}}$$

$$K(D) = \frac{1-2D}{1-D} \; if \; D \leq 0.5$$

$$K(D) = \frac{2D-1}{D} \; if \; D > 0.5$$

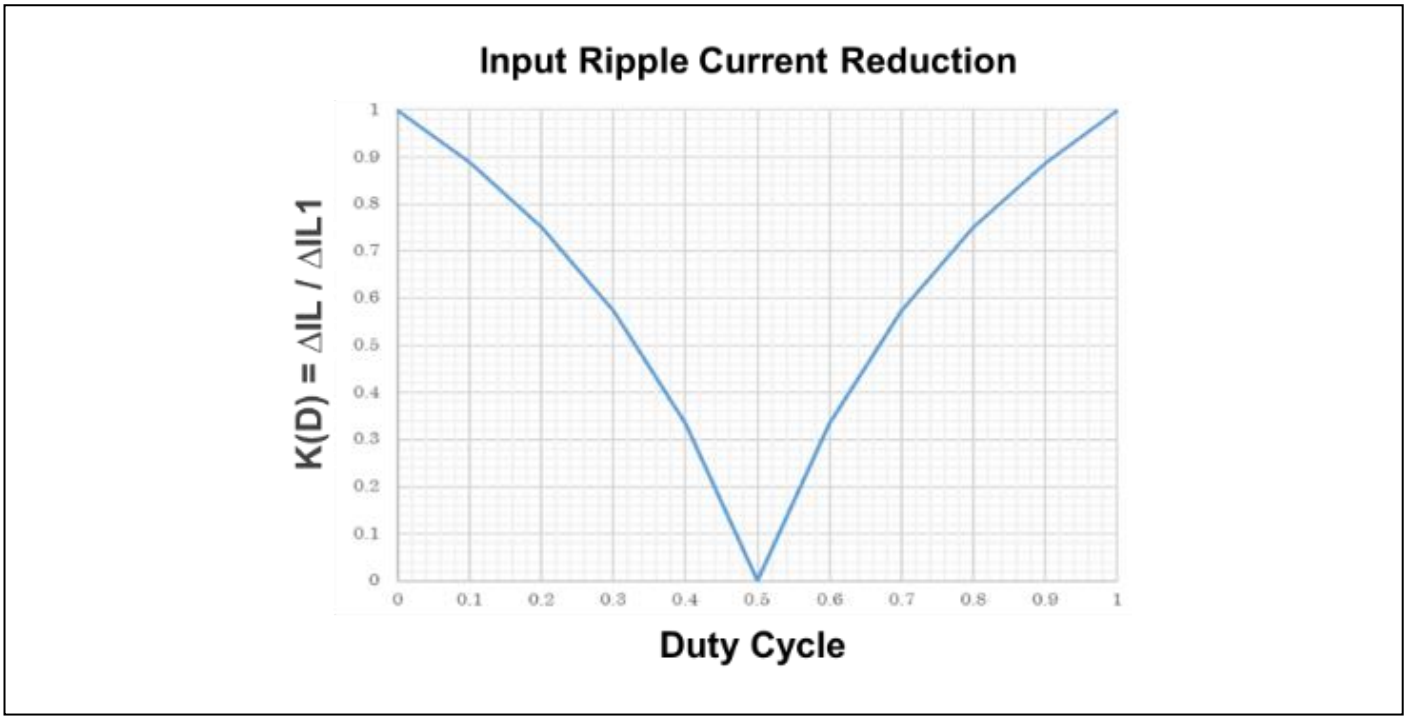


Figure 3. Input ripple current reduction.

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The inductor ripple current cancellation leads to the reduction in the magnetics volume. This is because reduction in energy is required due to the branching of inductors. The reduction is directly proportional to the number of branches.

Energy of a conventional boost inductor:

$$E = \frac{1}{2}LI^2$$

Two -phase inductor energy:

$$E_{2-phase} = \frac{1}{2}L\left(\frac{I^2}{2}\right) + \frac{1}{2}L\left(\frac{I^2}{2}\right) = \frac{1}{4}LI^2$$

The reduction in energy can lead up to 25% reduction in magnetic volume. From a single stage to 2-stage interleaved PFC.

3. IPFC Hardware Design

Input Section

3.1.1 EMI Section

The EMI filter section is a single stage filter to suppress both the Common mode (CM) and the Differential Mode (DM) noises. The X- caps suppresses the differential mode noise. The Y-caps and the common-mode choke suppress the common mode noise. An inductance value of 3.0 mH (7448062603 CMC choke from Wurth) is selected for this design.

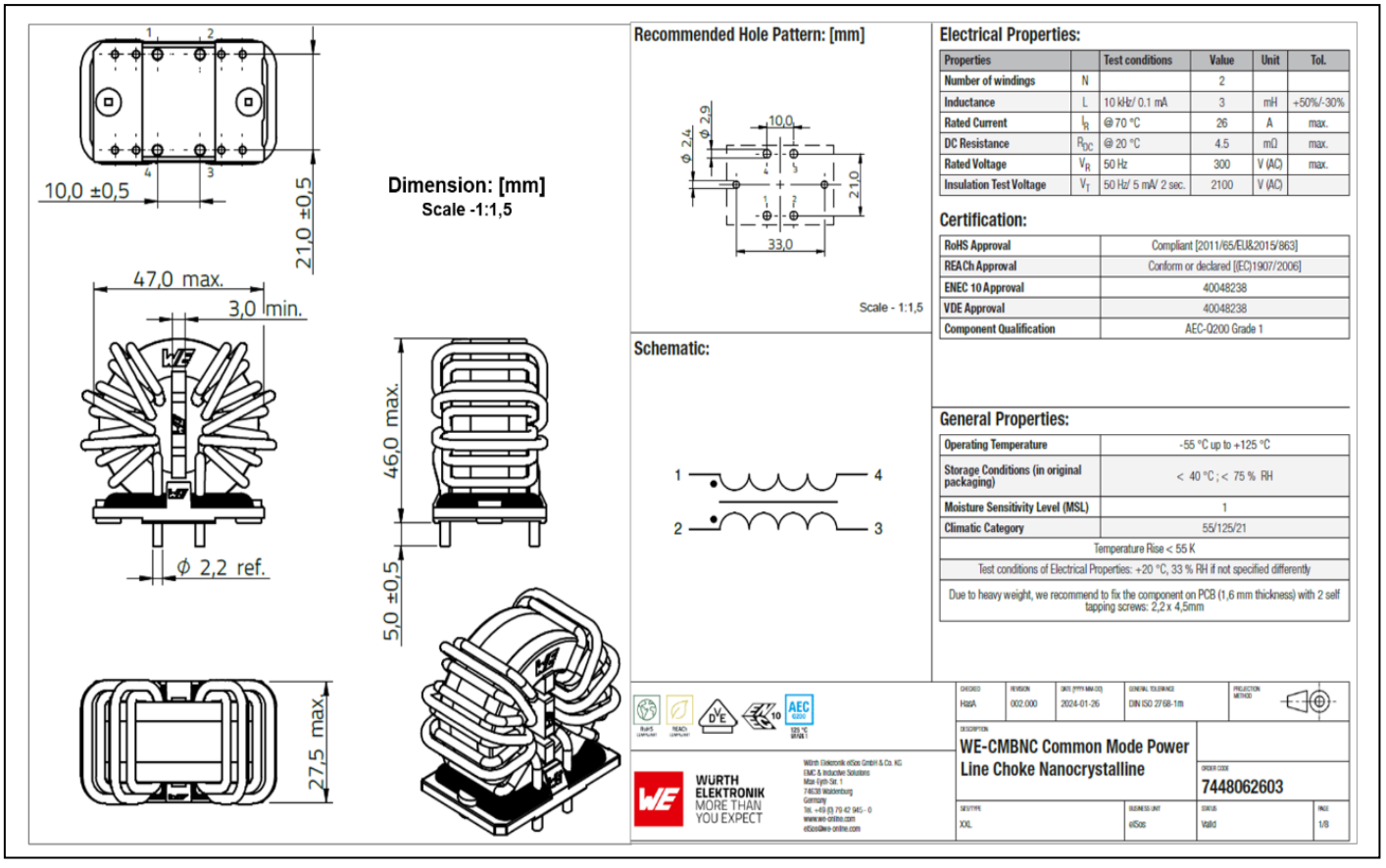


Figure 4. Common mode choke characteristics.

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3.1.2 Input Capacitor

The input capacitor is calculated based on the input current and the input voltage ripples. The input current ripple and the input voltage ripple considered for the calculation are 20% and 2% respectively. For obtaining the C_{in} value, we obtain the ripple current (ΔI) and the ripple voltage.

$$I_{ripple} = \Delta I_{ripple} \times I_{in_max}$$

$$= 0.2 \times \left(\frac{\sqrt{2} * P_{out}}{\eta * V_{in_min} * PF} \right)$$

$$= 0.2 \times \frac{\sqrt{2} * 3600}{.96 * 180 * .98} = 6.01\ A$$

$$V_{in_ripple} = \Delta V_{ripple} * V_{in_{\min(rect)}} = 0.02 * \left(\sqrt{2} * 180\right) = 5.09\ A$$

Input capacitance value:

$$C_{in} = \frac{I_{ripple}}{8 * f_{sw} * V_{in_ripple}}$$

$$= 2.27\ \mu F$$

A standard value 3.3 µF capacitor is selected to take into account the EMI performance into consideration.

3.1.3 Bridge Rectifier

Bridge rectifier has been selected based on the average input current and input voltage and considering all the losses at worst-case condition. The calculations are included as below:

Maximum input RMS current through the diode is:

$$I_{in_RMS} = \frac{P_{out}}{V_{in_min} * \eta}$$

$$= \frac{3600}{180 * .96} = 21.05\ A$$

Average current through diode is:

$$I_{D_avg} = \frac{\sqrt{2} * I_{in_RMS}}{\pi}$$

$$= \frac{\sqrt{2} * 21.05}{\pi} = 6.70\ A$$

To lower the total power dissipation, the bridge rectifier considered are with diodes with lower forward voltage drop. The bridge rectifier selected for this design is GBI40W from Diotec Semiconductor rated at 40 A. The loss component in the bridge is calculated as:

$$P_{bridge} = 2 * I_{D_{avg}} * V_f = 14.74\ \text{W}$$

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Boost Section

3.2.1 Boost Inductor design

The boost inductor stores energy during turn-on period and delivers the load during turn-off period. Here a powdered core is selected for lower core losses, high saturation flux density and good temperature performance. The inductances of the IPFC is calculated as:

$$L1 = L2 = \left(\frac{2*V_{ac-min}}{\Delta I* I_{L_avg}}\right) * \left(\frac{V_{out}-\sqrt{2}*V_{ac_min}}{V_{out}}\right) * \frac{1}{f_{sw}}$$

$$= \frac{2 * 180}{0.2 * 14.7} \times \frac{390 - \sqrt{2} * 180}{390} \times \frac{1}{65k} = \ 663.903\ \mu H$$

The detail of the inductor is included in **Table 1**.

Table 1. Inductor specification.

Specification	Value	Unit
PART #	NPS184075	
Inductance	680	µH
DCR	28.67	mΩ
A_L Value	171	nH
Number of turns (N= sqrt (L/AL))	63	turns
Wire Diameter	1.8 (15 SWG)	mm

3.2.2 Boost Diode

The selection of diode in a PFC circuit has major impact on the efficiency of the converter. Conventionally fast recovery diodes are selected as a boost diode as their reverse recovery characteristics make them affect the converter 's overall performance.

The reverse blocking voltage is usually selected based on maximum output voltage. While the current rating of the diode is decided by employing the simple thumb rule of 1 A / 150 W. This leads to a current rating of 24A, and hence a diode rated at 30 A is selected. The average current in the diode is:

$$I_{D_avg} = \frac{P_{out}}{Vout} = \ \frac{3600}{390} = 9.23\ A$$

The conduction losses in the diode are:

$$P_{cond} = I_{D_avg} * \ V_f = \ 9.23 * 1.53 = 14.12\ A$$

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Output Capacitor

Output capacitor also known as bulk capacitor is usually designed on the minimum hold-up time (t_{min}) in case of a power outage. Here, t_{min} is considered as 10ms, the duration of one-half cycle of the ac-line.

$$C_{out} \geq \frac{2 * P_{out} * t_{min}}{V_{out}^2 - V_{out_min}^2}$$

$$= \frac{2 * 3600 * 10m}{390^2 - 350^2} \geq 2676.57\ \mu F$$

The selected configuration is 4680 µF, 450 V electrolytic capacitors.

Gate Driver Design

The two interleaved switches are driven by **ROHM's** BD2310G gate driver. The single channel driver is capable of driving either N-ch FET and IGBT. The driver has a wide gate drive voltage range 4.5 V-18 V. It also has an added feature of built in UVLO function. The package of SSOP-05 makes the IC more compact.

Auxiliary Power Supply (AuxPS) Design

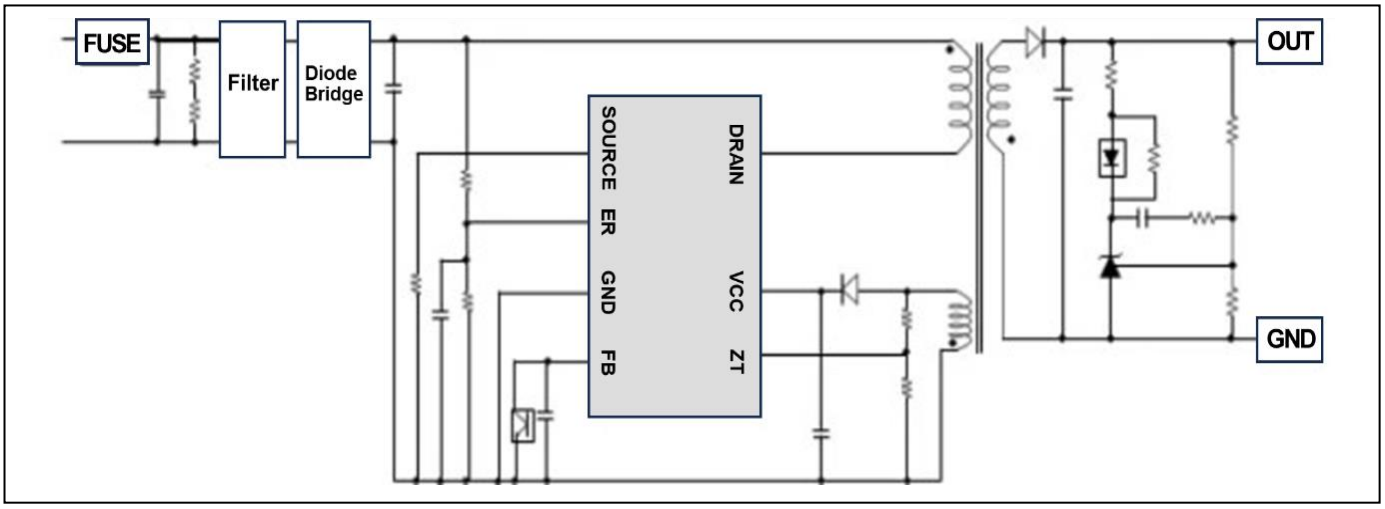


Figure 5. Auxiliary Power Supply (**AuxPS**).

The Auxiliary Power Supply (**AuxPS**) powers up the gate drivers, controllers and op-amps that are used across different sections in the converter. The chosen part for Aux-PS is **ROHM's** BM2P063HK series. This is a PWM type DC/DC converter which supports both isolated and non-isolated devices, enabling simpler design of various types of low power consumption electrical converters. This series also has a built-in starter circuit that can withstand up to 800V, thereby contributing to lower power consumption. The switching frequency is fixed at 65kHz, 100k Hz or 130 kHz. At light load, the switching frequency is reduced and high efficiency can be achieved. A frequency hopping function is also built-in that contributes to lower EMI. In addition, this product has a built-in super junction MOSFET rated at 800 V. The flyback topology has been implemented with 3 outputs i.e. 15 V, 12 V and 5 V. The +15 V and +5 V are further connected to LDO's for realizing +3.3 V outputs used for the controller and op-amps. +12 V feeds the PFC driver at the primary side.

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4. Control Loop

The equivalent block diagram showing the control loop structure is shown in **Figure 6.**

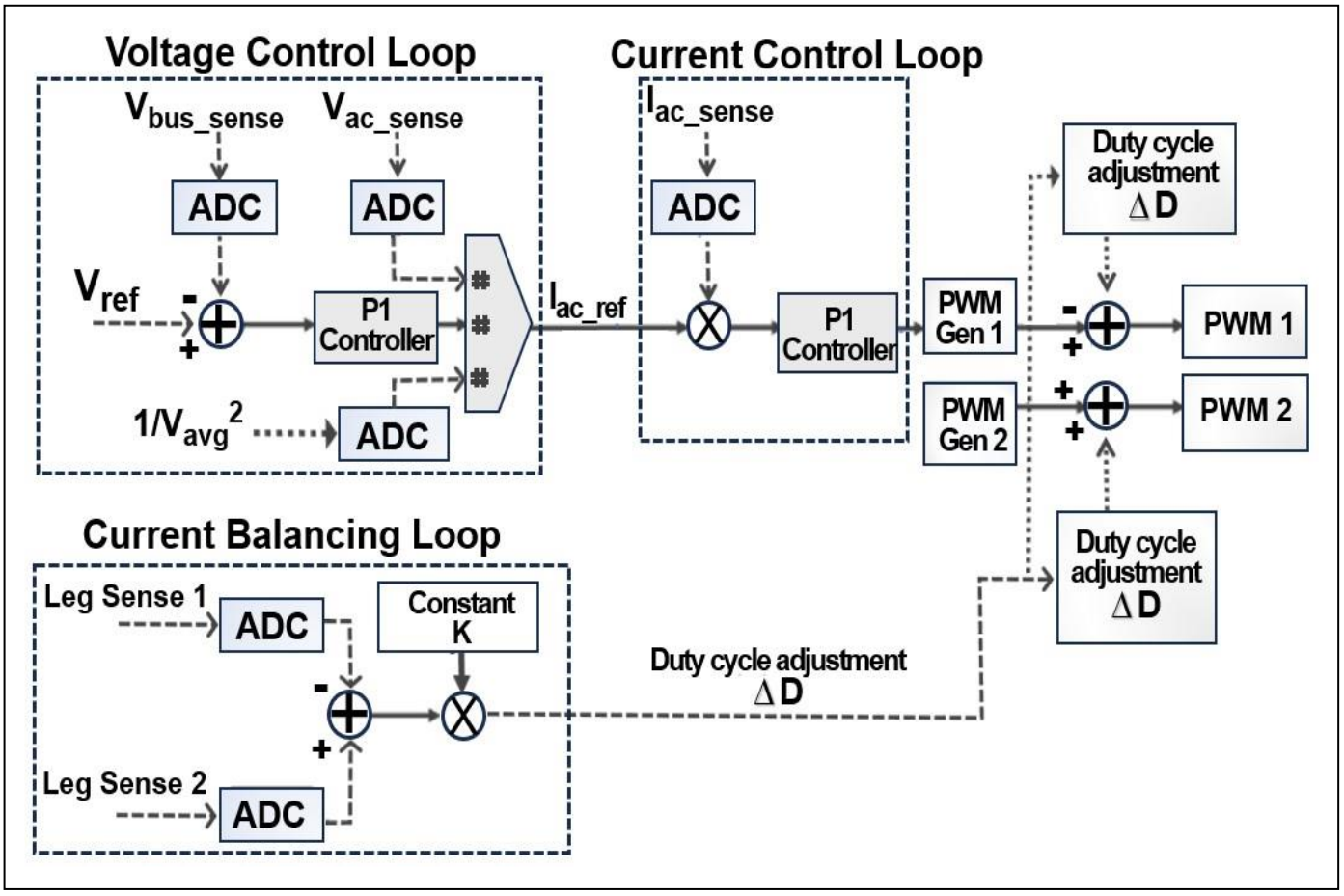


Figure 6. Control loop structure.

The current loop uses average current mode control for achieving power factor correction. In this control the scheme matches the average input current to the reference. The control operates in the Continuous Conduction Mode (**CCM**) across all levels depending on the inductor values and provides several benefits such as.

- Higher degree of accuracy
- Suitable for higher power levels
- No need of slope compensation
- Good noise immunity

The control structure can be divided into 3 main loops:

- Voltage Control Loop
- Current Control Loop
- Current Balancing Loop

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Voltage Control Loop

The voltage loop constitutes the outermost loop of the control structure. This loop takes care of regulating the output voltage across all loads and supply conditions. The actual DC bus voltage (V_{bus_sense}) and the reference bus voltage (V_{ref}) are compared. The difference between the reference value gives the error value. The error signal is fed to a PI controller.

The output of the PI controller is the input to the current control loop, which acts as the reference (I_{ac_ref}). Usually, the voltage loop is operated at a very low rate (~ 1/10 times) when compared to the current loop and has a smaller bandwidth. The bandwidth is selected considering the effect of input frequency ripple on the bus voltage.

Current Control Loop

The current control loop is the inner loop of the control structure that decides the duty of the switches. The reference signal from the voltage loop is compared with the input current sense (I_{ac_sense}). The error signal is fed to a PI controller. This corrects the error by forcing the Input current to follow the reference current, which takes care of the essential operation of the PFC.

The current control loop needs to operate at a relative higher frequency and higher bandwidth when compared to the voltage control loop, The current loop requires to sample the rectified AC signal at twice the input frequency and is operated at a frequency that is a multiple of the switching frequency.

Current Balance Loop

As explained in the previous section, interleaved topology adds several advantages. However, as the control structure of the loop gets affected to accommodate additional current balance and ensure that the switches share the load equally. Due to the variation of multiple parameters across both the switches, there is also a drift in the values measured across each leg. If same duty is applied to both the switches, it may lead to over-stressing a particular leg causing thermal stress. This may lead to triggering of protection functions. Hence, a current balancing loop becomes necessary.

Here, a cycle-by-cycle duty ratio adjustment method is adopted for current balancing. The current across each leg is sensed and then compared with each other. The error determines the imbalance in the legs. If the first leg current is higher, the error is positive and negative for vice-versa. The multiplier increases / decreases the duty cycle based on the polarity of the error.

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5. Experimental Results

Performance

Performance readings are captured using a HIOKI Power Analyzer PW6001. The measurements are taken for Si FRD & SiC SBD at 120 Vac and 230 Vac. The load is derated to half of the full load at low-line.

5.1.1 Performance at 120 V_{ac}

Table 2. Performance parameters at 120 Vac.

P_{out_load} (%)	V_{in_AC} (V)	I_{in_AC} (A)	PF	P_{in} (W)	V_{out} (V)	I_{out} (A)	P_{out} (W)	η (%)
10	121.885	1.679	0.9727	199.069	377.095	0.4948	186.587	93.73
20	121.883	3.217	0.9752	382.436	377.437	0.960	362.453	94.77
30	121.871	4.726	0.9882	569.201	377.858	1.4277	539.468	94.78
40	121.857	6.331	0.9931	766.111	377.810	1.9257	727.549	94.97
50	121.844	7.882	0.9952	955.675	377.960	2.3991	906.764	94.88
60	121.832	9.431	0.9963	1144.725	377.840	2.8716	1085.005	94.78
70	121.818	10.973	0.9969	1332.654	377.411	3.3409	1260.892	94.62
80	121.805	12.570	0.9974	1527.035	377.952	3.8173	1442.756	94.48
90	121.791	14.146	0.9977	1718.782	378.112	4.2856	1620.437	94.28
100	121.778	15.828	0.9979	1923.445	378.107	4.7781	1806.633	93.93

5.2.2 Performance at 230 Vac

Table 3. Performance parameters at 230 Vac.

P_{out_load} (%)	V_{in_AC} (V)	I_{in_AC} (A)	PF	P_{in} (W)	V_{out} (V)	I_{out} (A)	P_{out} (W)	η (%)
10	229.79	1.75	0.9201	371.04	379.36	0.94	358.42	96.60
20	229.79	3.38	0.9554	741.50	379.86	1.90	720.44	97.16
30	229.78	5.01	0.9677	1113.41	379.98	2.85	1082.38	97.21
40	229.77	6.62	0.9787	1488.02	380.05	3.81	1449.21	97.39
50	229.76	8.22	0.9857	1861.82	379.66	4.77	1811.97	97.32
60	229.75	9.77	0.9899	2221.47	379.90	5.69	2161.44	97.30
70	229.73	11.40	0.9927	2599.56	379.77	6.65	2526.70	97.20
80	229.72	12.99	0.9944	2968.21	379.69	7.59	2882.01	97.10
90	229.71	14.61	0.9956	3342.37	379.95	8.54	3243.31	97.04
100	229.69	16.24	0.9965	3718.17	380.12	9.47	3601.26	96.86

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The improved efficiency of the IPFC with Si FRD & SiC SBD can be seen in the **Figure 7** and **Figure 8**. Clearly, the IPFC achieves higher efficiency with SiC SBD. As discussed in **Section 3**, the SiC SBDs exhibit a superior performance when compared to the Si FRD counterparts.

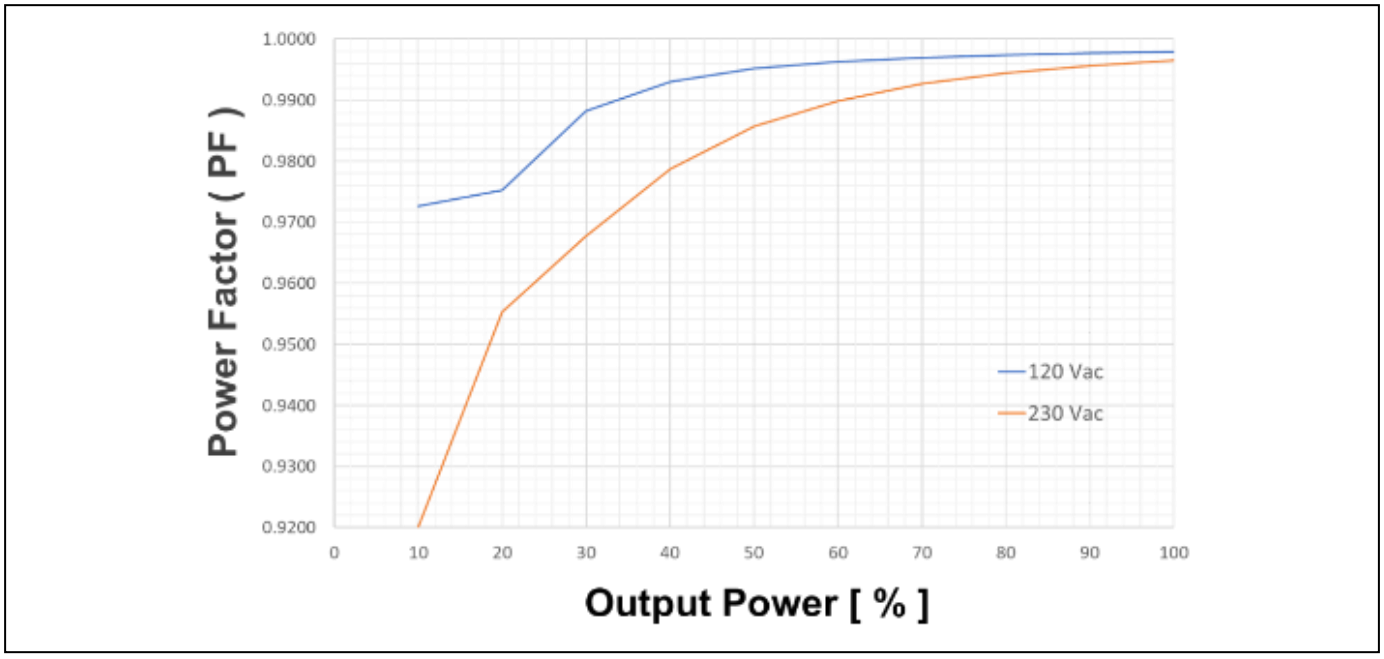


Figure 7. Power Factor Versus Output Power.

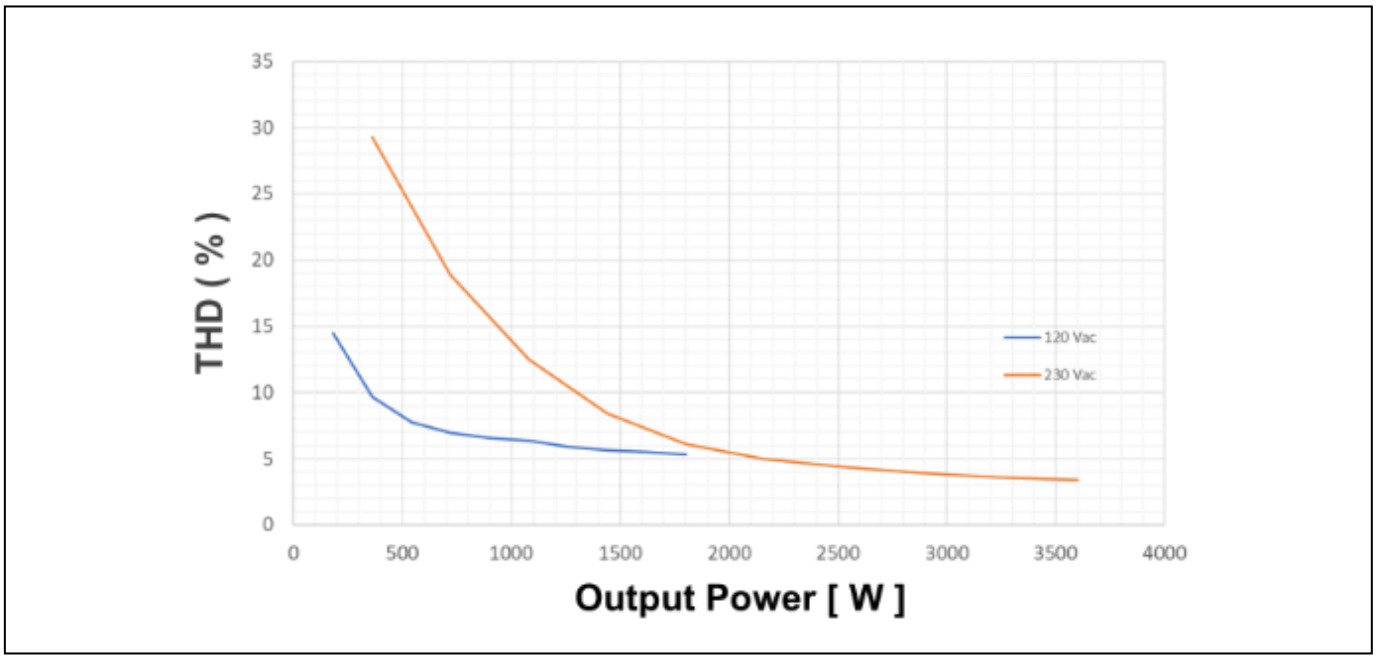


Figure 8. THD Versus Output Power.

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6. PCB Layout Layer Stack Up

Power Board Layer Stack				
Layer	Construction	Copper Thickness	Dielectric Thickness	Part #
		Oz	mm	
	Solder Mask		0.02	EMP 110/5158, DK4.2
L1	Copper	1	0.07	
	Core		1.6	ISOLA 185HR
L2	Copper	1	0.07	
	Solder Mask		0.02	EMP 110/5158, DK4.2
Total Thickness:			1.78	±10%

Auxiliary Power ssssupply Board Layer Stack				
Layer	Construction	Copper Thickness	Dielectric Thickness	Part #
		Oz	mm	
	Solder Mask		0.02	EMP 110/5158, DK4.2
L1	Copper	1	0.035	
	Core		1.6	ISOLA 185HR
L2	Copper	1	0.035	
	Solder Mask		0.02	EMP 110/5158, DK4.2
Total Thickness:			1.71	±10%

MCU Board Layer Stack				
Layer	Construction	Copper Thickness	Dielectric Thickness	Part #
	Solder Mask		0.02	EMP 110/5158, DK4.2
L1	Copper	1	0.035	
	Core			ISOLA 185HR
L2	Copper	1	0.035	
	Core			ISOLA 185HR
L3	Copper	1	0.035	
	Core			ISOLA 185HR
L4	Copper	1	0.035	
	Solder Mask		0.02	EMP 110/5158, DK4.2
Total Thickness:			1.0	±10%

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7. Schematics

IPFC section

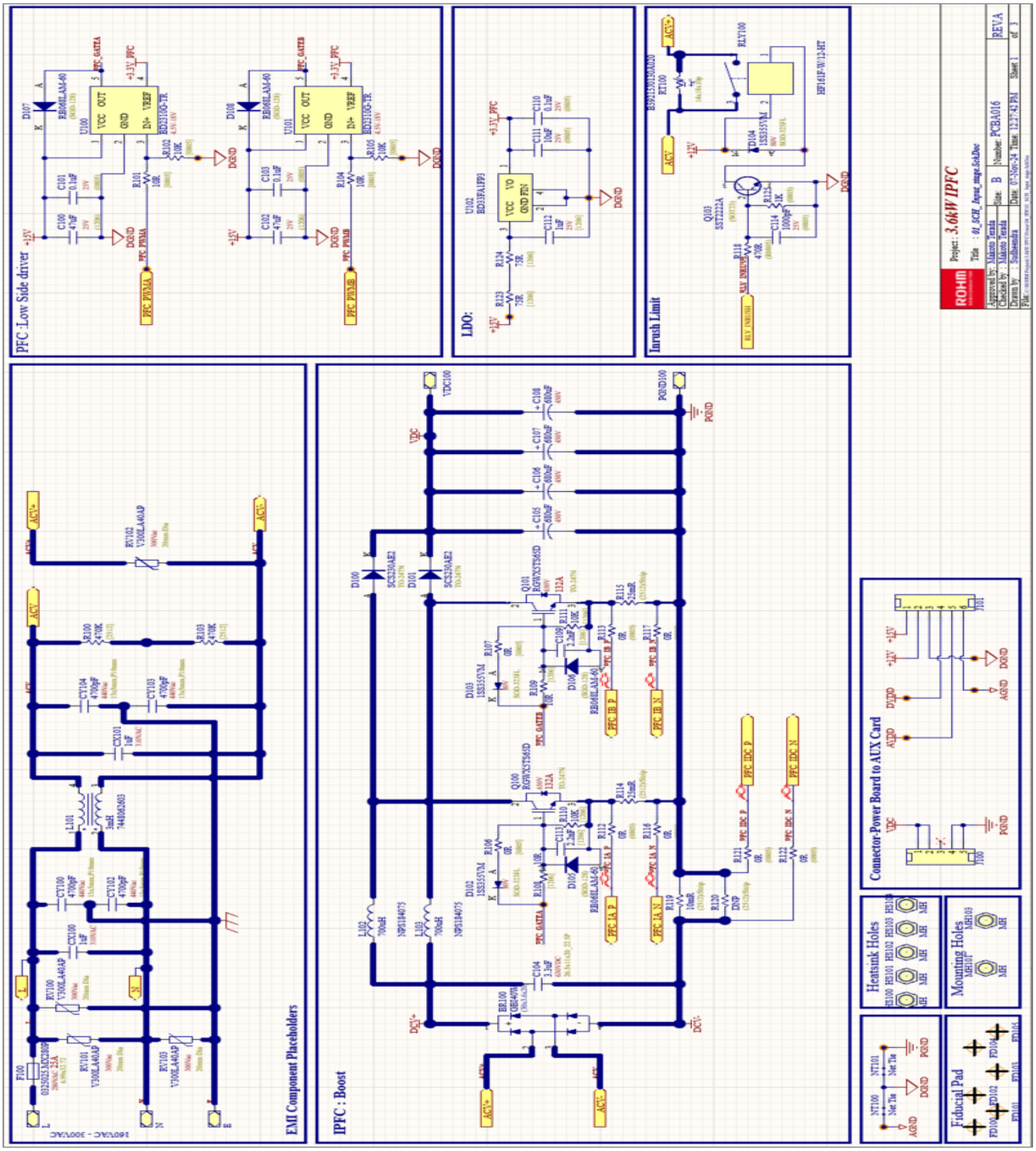


Figure 9. Power Board schematics.

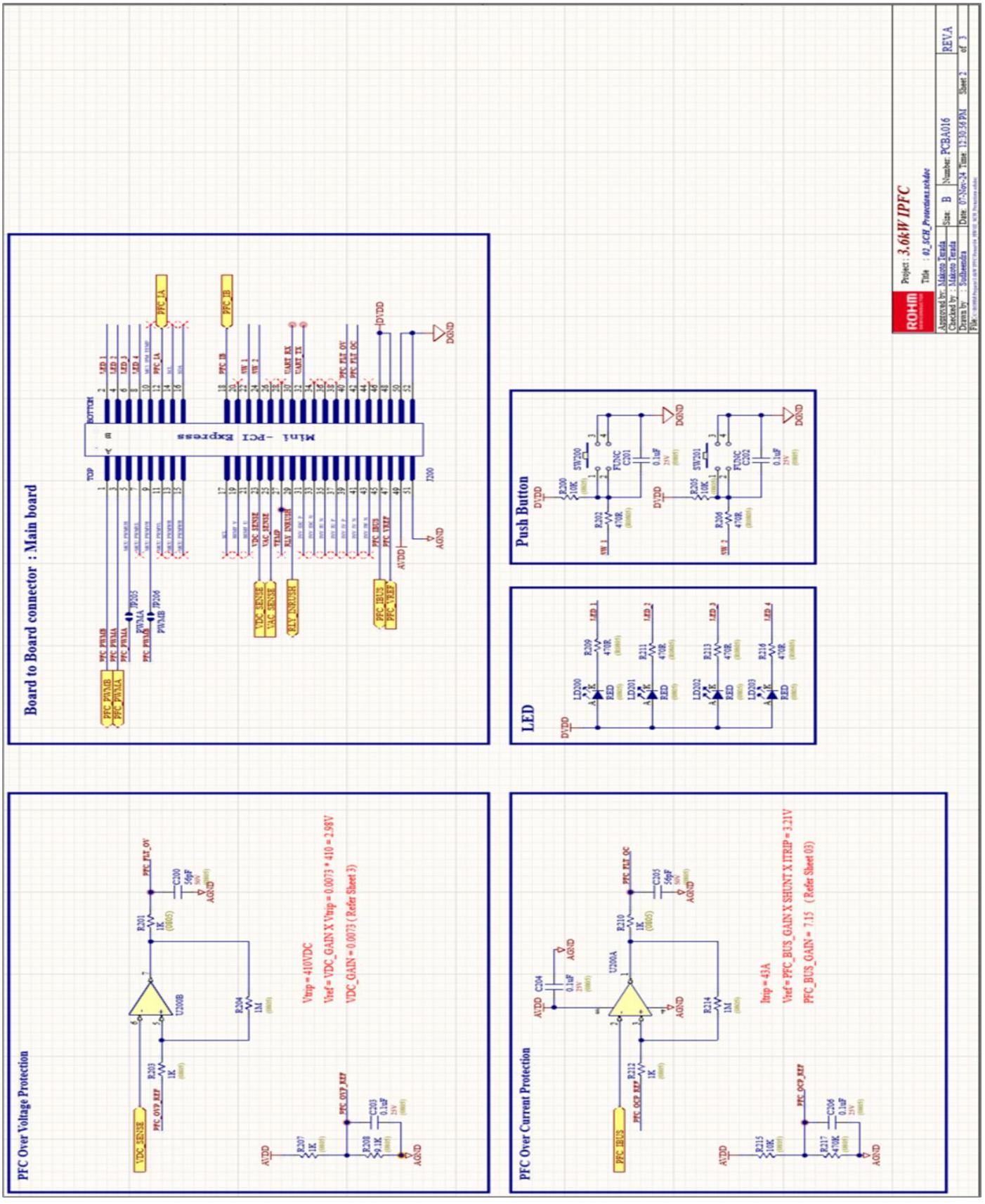


Figure 10. Power Board connector schematics.

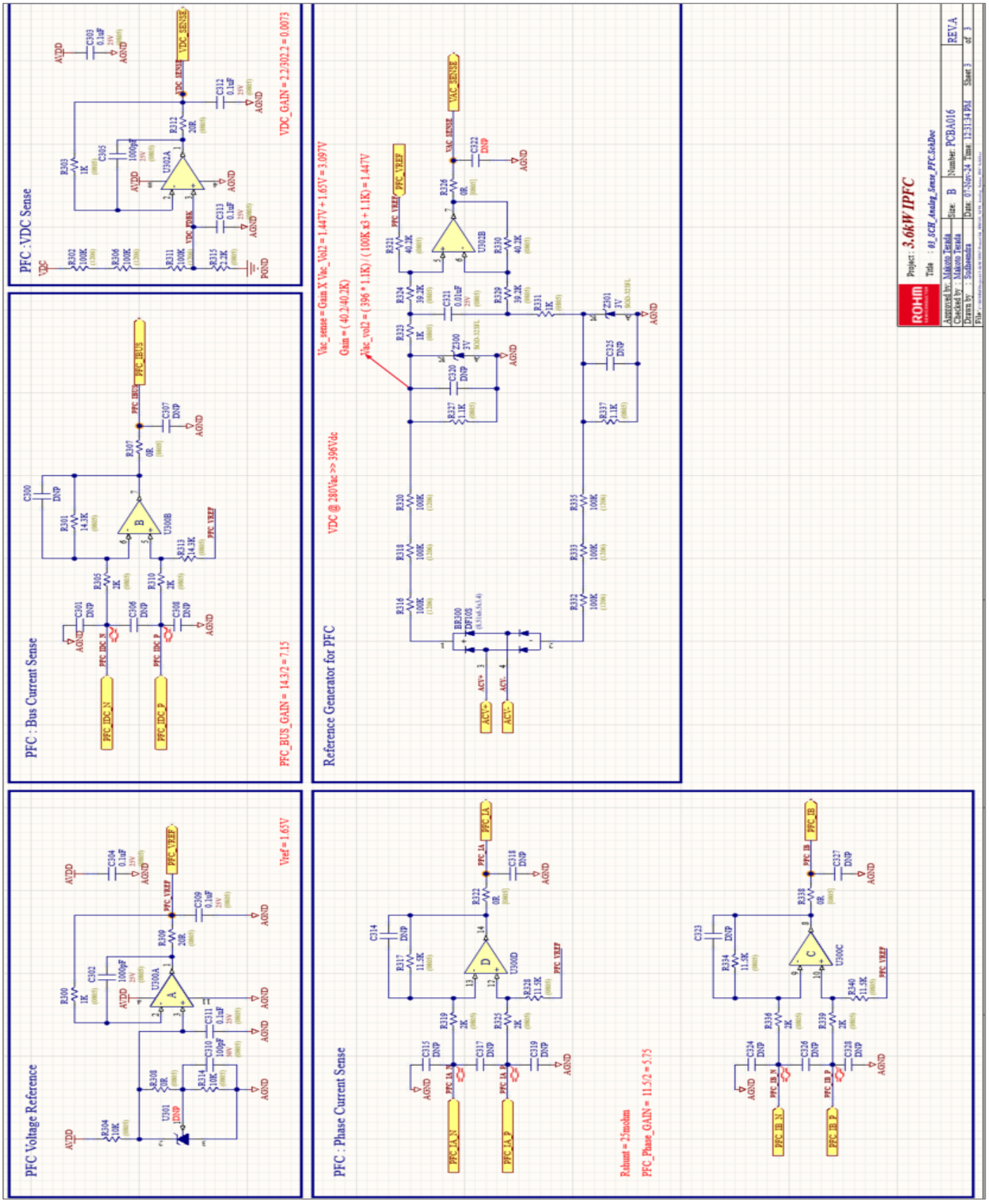


Figure 11. Power Board analog sense schematics.

Auxiliary Power Supply (AuxPS) section

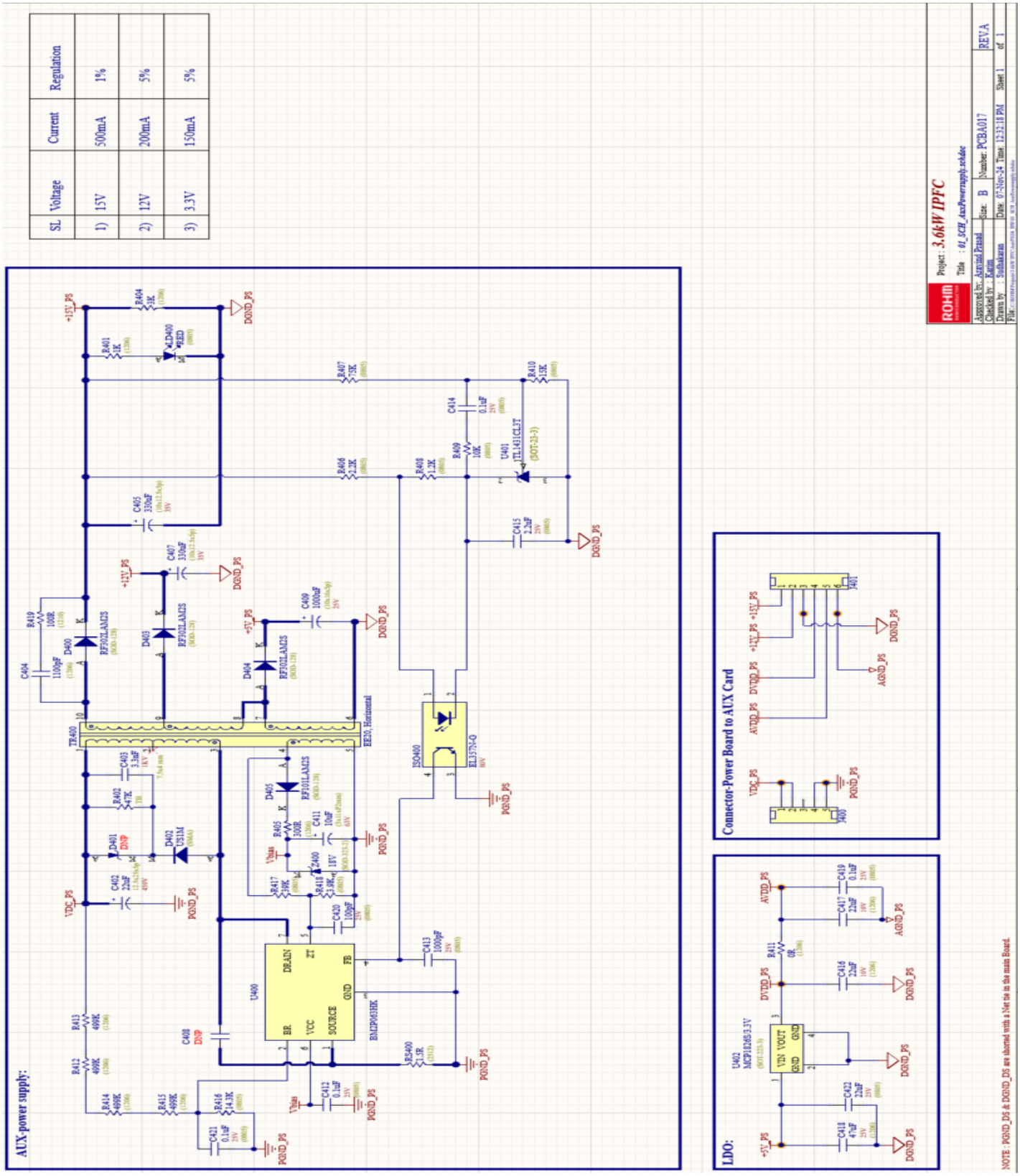


Figure 12.. Auxiliary power supply schematics.

Controller Section

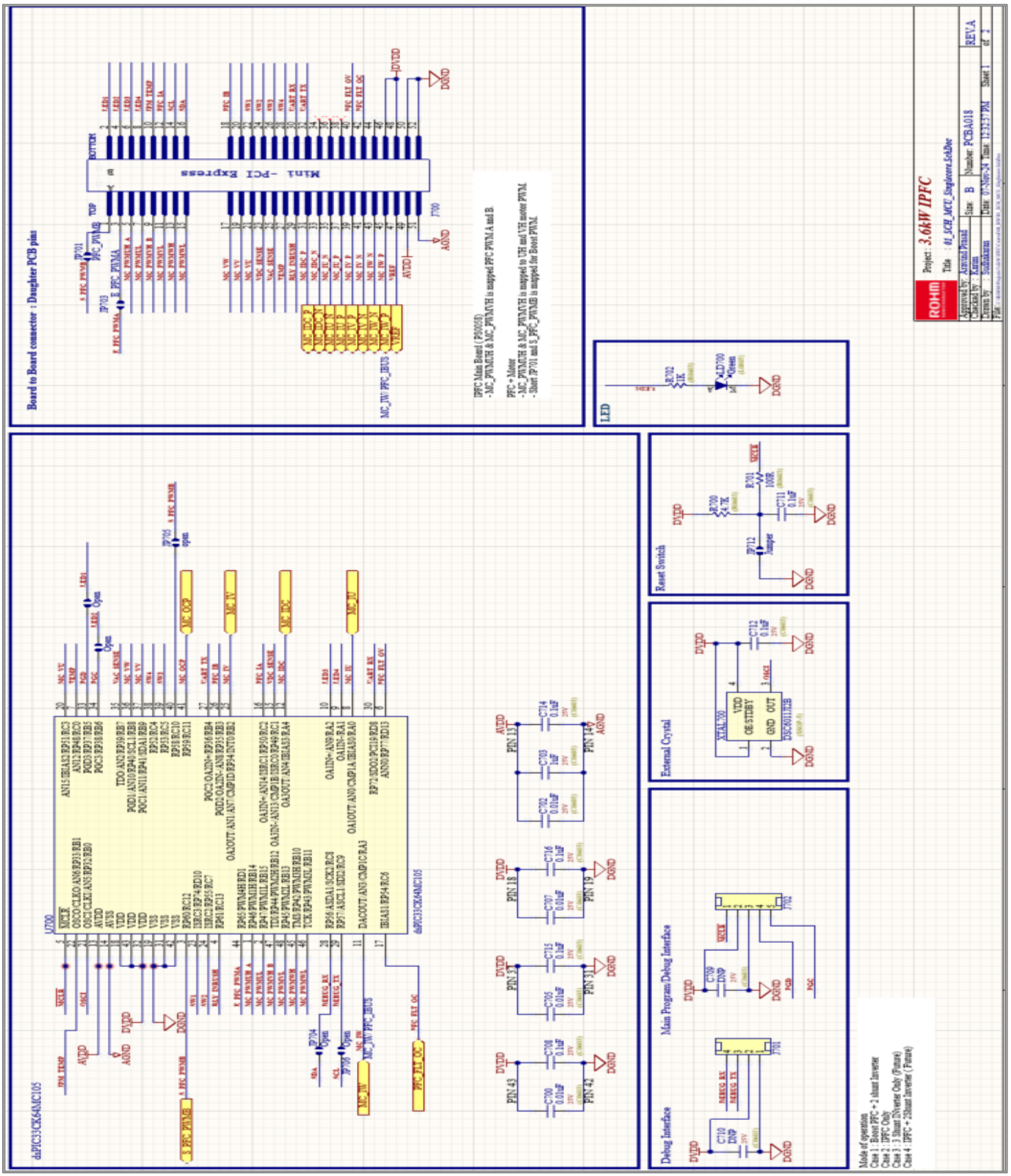


Figure 13. MCU schematics.

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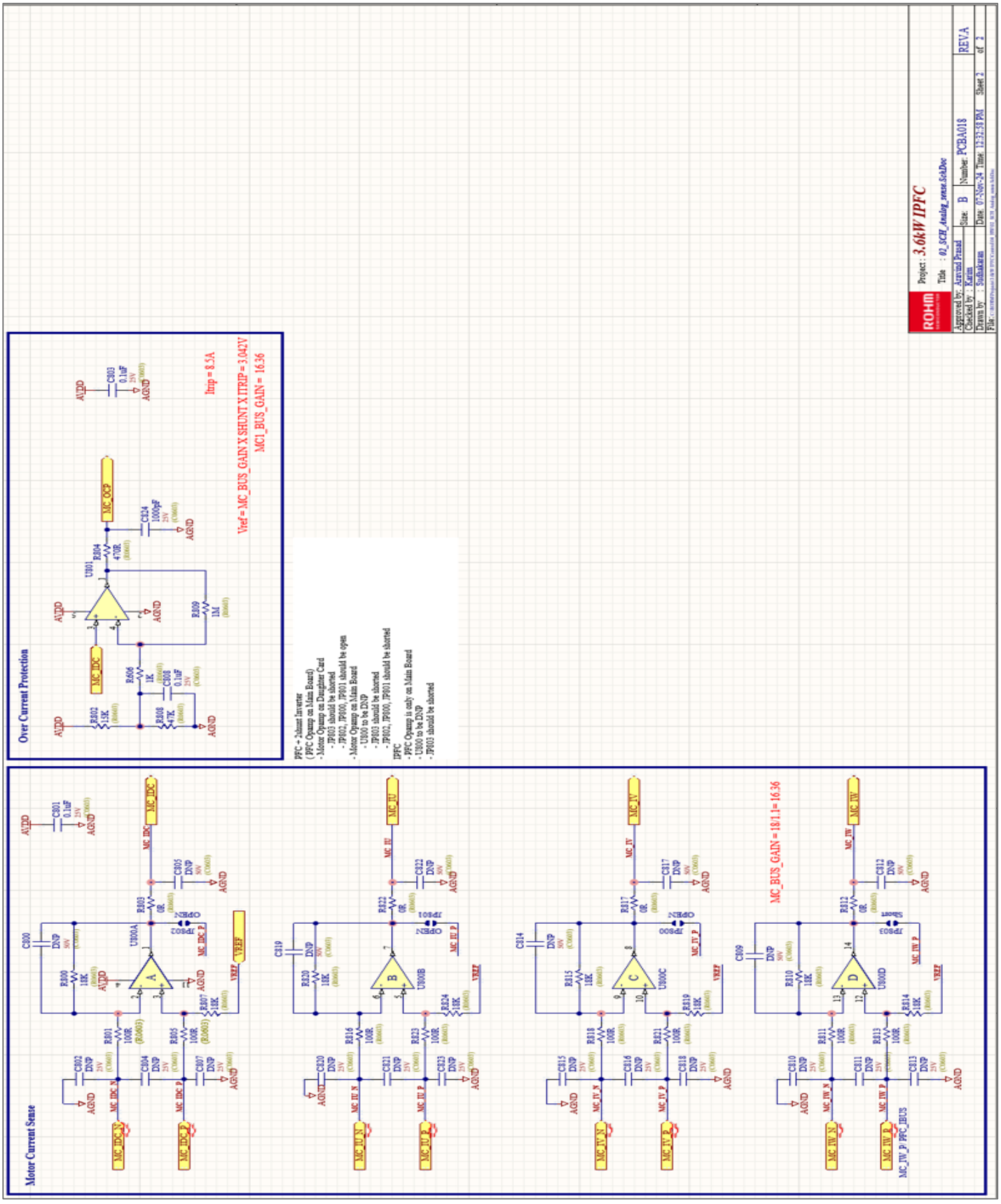


Figure 14. Motor current sense schematics.

8. Bill of materials (BOM)

Interleaved Boost PFC (IPFC) BOM

Designator	Comment	PART Description	Package	Manufacturer	Part Number
BR100	GBI40W	Diode, TH, GBI40W,1600V, 40A	(30x3.6x20)	Diodes tec	GBI40W
BR300	DF10S	Diode,SMD,,DF10S,1000V	(8.51x6.5x3.4)	Diodes Inc	DF10S
C100, C102	47uF	CAP NP, SMD, Ceramic, 47uF, 25V, 20%, X5R	(1206)	TDK	C3216X5R1E476M160AC
C101, C103, C110, C201, C202, C203, C204, C206, C303, C304, C309, C311, C312, C313	0.1uF	CAP NP, SMD, Ceramic, 0.1uF, 25V, 10%, Y5V	(0805)	Yageo	CC0805ZRY5V8BB104
C104	3.3uF	CAP NP, TH, Ceramic, 3.3uF, 630VDC, 1.0%, X5R	(26.5x11x20_22.5P)	Kemet	R71PN433050H6K
C105, C106, C107, C108	680uF	CAP POL, TH, Electrolytic, 680uF, 450V, 10%, 2.15A,105°C	(35X60X10p)	Kemet	ELD687M450AT5AA
C109, C113	2.2nF	CAP NP, SMD, Ceramic, 2.2nF, 50V,10%, X7R	(1206)	Yageo	CC1206KPX7R9BB222
C111	10uF	CAP NP, SMD, Ceramic,10uF, 25V, 10%, X5R	(0805)	Yageo	CC0805KKX5R8BB106
C112	1uF	CAP NP, SMD, Ceramic,1uF, 25V, 10%, X7R	(1206)	Yageo	AC1206KKX7R8BB105
C114, C302, C305	1000pF	CAP NP, SMD, Ceramic, 1000pF, 25V,10%, X7R	(0805)	Yageo	CC0805KRX7R8BB102
C200, C205	56pF	CAP NP, SMD, Ceramic, 56pF, 50V,5%, C0G	(0805)	Yageo	AC0805JRNPO9BN560
C310	100pF	CAP NP, SMD, Ceramic,100pF, 50V, 10%, X7R	(0805)	Yageo	CC0805KRX7R9BB101
C321	0.01uF	CAP NP, SMD, Ceramic, 0.01uF, 25V, 10%, X7R	(0805)	Yageo	CC0805KRX7R8BB103
CX100, CX101	1uF	CAP NP,TH,Ceramic,1uF, 310VAC	(TH_26.5x11x20mm_22.5P)	Desai Electronics	MPB X2
CY100, CY102, CY103, CY104	4700pF	CAP NP, TH, Ceramic, 4700pF, 440Vac, 20%,Y5U	(13x5mm,P10mm)	Kemet	C961U472MZWDBA7317
D100, D101	SCS230AE2	SiC Schottky Diode, RECT, 650V,30A	(TO-247s)	**ROHM**	SCS230AE2
D102, D103, D104	1SS355VM	Diode, SMD, SIGNAL,1SS355VM, 80V,100mA −55°C ~ 150°C	SOD-323 FL	**ROHM**	1SS355VMTE-17
D105, D106	RB068LAM-60	Diode, SMD, RECT, RB068LAM-60, 60V	(SOD-128)	**ROHM**	RB068LAM-60
F100	0325025.MX280P	FUSE, TH, 0325025.MX280P, 280VAC	(6.99x32.72)	Little Fuse	0324025.MX280P
HS100, HS101, HS102, HS103, HS104, MH101, MH103	MH	Hole, TH, MH			
J100	CON_HDR_5	Connector, TH, 5 positions	(2.54mm Pitch)	Wurth	61300511021
J101	CON_HDR_6	Connector, TH, 6 positions	(2.54mm Pitch)	Wurth	61300611021

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J200	DIP 5.08x3N	Connector, PCI Express, DIP 5.08x3N	(SMD-52POS)	TE-Connectivity	1759546-1
L101	3mH	Common Mode Choke, TH, 3mH, 250V AC, 4.5mOHM	(47x27mm)	Wurth	7448062603
L102, L103	700uH	Inductor, TH, 700uH, 250VAC, 1%	(50X22mm)	poco Holding	NPS184075
LD200, LD201, LD202, LD203	RED	Diode, SMD, RED, 30mA75mW	(0805)	**ROHM**	SML-M13VTT86
Q100, Q101	RGWX5TS65D	IGBT, TH, N Ch COPACK,RGW00TS65D, 650V, 75A, -40°C ~ 175°C (TJ)	(TO-247N)	**ROHM**	RGWX5TS65D
Q103	SST2222A	Transistor, SMD, NPN,SST2222A, 40V, 0.6A–55°C ~ 150°C	(SOT23)	**ROHM**	SST2222A
R100, R103	470K	Resistor, SMD, 470K, 1%, 1 W	(2512)	Yageo	AA2512FK-07470KL
R101, R104	10R	Resistor, SMD,10R,1%	(0805)	**ROHM**	ESR10EZPF10R0
R102, R105	10K	Resistor, SMD,10K,150V, 1%, 1/8 W	(0805)	**ROHM**	ESR10EZPF1002
R106, R107, R307, R322, R326, R338	0R	Resistor, SMD, 0R, 1%, 1/8 W	(0805)	**ROHM**	ESR10EZPJ000
R108, R109	10R	Resistor, SMD, 10R, 1%, 1/4 W	(1206)	**ROHM**	SFR18EZPJ100
R110, R111	10K	Resistor, SMD, 10K, 150V,1%, 1/4 W	(1206	**ROHM**	SFR18EZPF1002
R112, R113, R116, R117, R121, R122, R308	0R	Resistor, SMD, 0R, 1%, 400mW	(0805)	**ROHM**	ESR10EZPF0000
R114, R115, R119, R120	10mR	Resistor, SMD, 10mR, 1%, 3W	(2512)/Strip	**ROHM**	GMR100HTBFA10L0
R118, R202, R206, R209, R211, R213, R216	470R	Resistor, SMD, 470R,-,1%,1/8W	(R0805)	**ROHM**	KTR10EZPF4700
R123, R124	75R	Resistor, SMD, 75R,1%, 1/2W	(1206)	**ROHM**	ESR18EZPF75R0
R125, R201, R203, R207, R210, R212, R300, R303, R323, R331	1K	Resistor, SMD,1K,1%, 400mW	(0805)	**ROHM**	ESR10EZPF1001
R200, R205, R215, R304, R314	10K	Resistor, SMD, 10K,1%, 1/8 W	(0805)	**ROHM**	MCR10EZPF1002
R204, R214	1M	Resistor, SMD,1 M,1%, 400mW	(0805)	**ROHM**	ESR10EZPF1004
R208	9.1K	Resistor,SMD,9.1K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF9101
R217	470K	Resistor, SMD, 470K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF4703
R301, R313	14.3K	Resistor, SMD, 14.3K, 1%,1/8 W	(0805)	**ROHM**	KTR10EZPF1432
R302, R306, R311, R316, R318, R320, R332, R333, R335	100K	Resistor, SMD, 100K, 1%, 1/4 W	(1206)	**ROHM**	KTR18EZPF1003
R305, R310, R319, R325, R336, R339	2K	Resistor, SMD, 2K,1%, 400 mW	(0805)	**ROHM**	ESR10EZPF2001
R309, R312	20R	Resistor, SMD, 20R,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF20R0

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R315	2.2K	Resistor, SMD,2.2K,1%,1/8 W	(0805)	**ROHM**	KTR10EZPF2201
R317, R328, R334, R340	11.5K	Resistor,SMD,11.5K,1%,1/8 W	(0805)	**ROHM**	KTR10EZPF1152
R321, R330	40.2K	Resistor, SMD, 40.2K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF4022
R324, R329	39.2K	Resistor, SMD, 39.2K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF3922
R327, R337	1.1K	Resistor, SMD, 1.1K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF1101
RLY100	HF161F-W/12-HT	Relay,TH,HF161F-W/12-HT, 12V, 10%, -40°C ~ 85°C	(30.4x15.9x23.3)	HONGFA	HF161F-W/12-HT
RT100	B59215J0130A020	PTC,TH, B59215J0130A020, 400 VDC,1%	(14x18x10p)	TDK	B59215J0130A020
RV100, RV101, RV102, RV103	V300LA40AP	Varistor, TH, V300LA40AP, 300Vac,	(20mm Dia)	Little Fuse	V300LA40AP
SW200, SW201	FUNC	Switch, TH, 4 Pin, FUNC	(6.2x6.2x3.5mm)	Wurth	4.30481E+11
U100, U101	BD2310G-TR	lLow Side GATE Driver, SMD, IC, BD2310G-TR, 4.5V-18V, 4A	(SSOP-5)	**ROHM**	BD2310G-TR
U102	BD33FA1FP3	LDO, SMD, IC, BD33FA1FP3, 3.3V, 100mA	(SOT-893K)	**ROHM**	BD33FA1FP3
U200	MCP6542-E/SN	Comparator, SMD, MCP6542-E/SN, 1.6 V ~ 5.5 V	(SOIC-8)	Microchip	MCP6542-E/SN
U300	MCP6024-I/ST	OpAmp, SMD, Op AMP Quad, MCP6024-I/ST, 5V	(TSSOP-14)	Microchip	MCP6024-I/ST
U301	DNP	Low Voltage Precision Adjustable Shunt Regulator, SMD, DNP	(SOT-23-3)		
U302	MCP6022T-E/SN	OpAmp, SMD, MCP6022T- E/SN, 1.6 V ~ 5.5 V	(SOIC-8)	Microchip	MCP6022T-E/SN
Z300, Z301	3V	Diode, SMD, ZENER, 3V, 200mW, -55°C ~ 150°C	(SOD-323FL)	**ROHM**	UDZVTE-173.0B

Controller BOM

Designator	Comment	PART Description	Package	Manufacturer	Part Number
C700, 702, C705, C707	0.01uF	CAP, P, SMD, Ceramic, 0.01uF, 25V, 5%, X7R	(C0603)	Yageo	AC0603JRX7R8BB103
C703	1uF	CAP, NP, SMD, Ceramic, 1uF, 25V, 10%, X5R	(C0603)	Yageo	CC0603KRX5R8BB105
C708, C711, C712, C714, C715, C716, C801, C803, C808	0.1uF	CAP, NP, SMD, Ceramic, 0.1uF, 25V, 10%, X7R	(C0603)	Yageo	CC0603KRX7R8BB104
C824	1000pF	CAP, NP, SMD, Ceramic, 1000pF, 25V, 10%, X7R	(C0603)	Yageo	CC0603KRX7R8BB102
J700	1759546-1	Connector, SMD, PCI Express, 1759546-1	(SMD-52)	TE-Connectivity	1759546-1
J701	CON_HDR_4	Connector, TH\,4 positions, CON_HDR_4	(2.54mm Pitch)	Wurth	61300411121

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J702	CON_HDR_5	Connector, TH\,5 positions, CON_HDR_5	(2.54mm Pitch)	Wurth	61300511121
LD700	Green	Diode, SMD, Yellow, Green, 20mA81mW	(L0805)	**ROHM**	SML-M13MTT86
R606, R702	1K	Resistor, SMD, 1K, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ102
R700	4.7K	Resistor, SMD, 4.7K, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ472
R701, R801, R805, R811, R813, R816, R818, R821, R823	100R	Resistor, SMD,100R, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ1000
R800, R807, R810, R814, R815, R819, R820, R824	18K	Resistor, SMD, 18K, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ183
R802	15K	Resistor, SMD, 15K, 5%, 300 mW	(R0603)	**ROHM**	ESR03EZPJ153
R803, R812, R817, R822	0R	Resistor, SMD ,0R, 5%,1/8 W	(R0603)	**ROHM**	SFR03EZPJ000
R804	470R	Resistor, SMD, 470R, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ471
R808	47K	Resistor, SMD, 47K, 5%, 300 mW	(R0603)	**ROHM**	SDR03EZPJ473
R809	1M	Resistor, SMD,1M, 5%, 1/4 W	(R0603)	**ROHM**	ESR03EZPJ105
U700	dsPIC33CK64MC105	MCU, SMD, MCU, dsPIC33CK64MC105, 3.3V	(TQFP-48)	Microchip	dsPIC33CK64MC105
U800	MCP6024-I/ST	OpAmp, SMD, Op AMP Quad, MCP6024-I/ST	(TSSOP-14)	Microchip	MCP6024-I/ST
U801	MCP6541	Comparator, SMD, MCP6541, 1.6 V ~ 5.5 V	(SOT23-5)	Microchip	MCP6541T-I/OT
XTAL700	DSC6011JI2B	MEMS_OscillatorSMDICDSC6011JI2B-20ppm	(SSOP-5)	Microchip	DSC6011JI2B

Aus PS BOM

Designator	Comment	PART Description	Package	Manufacturer	Part Number
C402	22uF	CAP POL, TH, Electrolytic, 22uF, 450V, 20%, -105°C	(12.5x25x5p)	Nichicon	UCY2W220MHD
C403	3.3nF	CAP NP, TH, Ceramic, 3.3nF, 1KV, 5%, U2J	(7.5x4 mm)	Murata	RDE7U3A332J4M1H03A
C405, C407	330uF	CAP POL,TH, Electrolytic, 330uF, 35V, 20%, -105DegC	(10x12.5x5p)	Nichicon	UVZ1V331MPD1TD
C409	1000uF	CAP POL, TH, Electrolytic, 1000uF, 25V, 20%, -105°C	(10x1s6x5p)	Nichicon	UVY1E102MPD

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C411	10uF	CAP POL,TH, Electrolytic,10uF, 63V, 20%, -105°C	(5x11xP2mm)	Nichicon	UVZ1J100MDD
C412, 414, C419, C421	0.1uF	CAP NP, SMD, Ceramic, 0.1uF, 25V, 20%, X7R	(0805)	Yageo	CC0805MRX7R8BB104
C413	1000pF	CAP NP, SMD, Ceramic,1000pF, 25V, 10%, X7R	(0805)	Yageo	CC0805KRX7R8BB102
C415	2.2uF	CAP NP, SMD, Ceramic, 0.1uF, 25V,20%, X5R	(0805)	Yageo	CC0805KRX5R8BB225
C416, C417	22uF	CAP NP, SMD, Ceramic, 22 uF, 10V, 10%, X5R	(1206)	Yageo	CC1206KKX5R6BB226
C418	47uF	CAP NP, SMD, Ceramic, 47uF, 25V, 20%, X5R	(1206)	TDK	C3216X5R1E476M160AC
C420	100pF	CAP NP, SMD, Ceramic,100pF, 25V, 10%, X7R	(0805)	Yageo	CC0805KRX7R8BB101
C422	22uF	CAP NP, SMD, Ceramic, 22uF, 25V, 20%, X5R	(0805)	Yageo	CC0805MKX5R8BB226
D400, 403, D404	RF302LAM2S	Diode, SMD, RECT, RF302LAM2S, 200V, 3A	(SOD-128)	**ROHM**	RF302LAM2S
D402	US1M	Diode, SMD, RECT, US1M, 1000V, 1A	(SMA)	Diodes Inc	US1M-13-F
D405	RF101LAM2S	Diode, SMD, RECT, RF101LAM2S, 200V, 1A	(SOD-128)	**ROHM**	RF101LAM2S
ISO400	EL357N-G	Optoisolator, SMD, IC, EL357N-G, 80V	(TSSOP-4)	Everlight	EL357N-G
J400	CON_HDR_5	Connector, TH, 5 positions	(2.54mm Pitch)	Wurth	61300511021
J401	CON_HDR_6	Connector, TH, 6 positions	(2.54mm Pitch)	Wurth	61300611021
LD400	RED	Diode, SMD, RED, 30mA75mW	(0805)	**ROHM**	SML-M13VTT86
R401	1K	Resistor, SMD,1K, 1%, 1/4 W	(1206)	**ROHM**	KTR18EZPF1001
R402	47K	Resistor, TH, DNP, 5%, TH, 2W	(TH)	YAGEO	CFR2WSJT-52-47K
R404	3K	Resistor, SMD, 3K,1%,1/4 W	(1206)	**ROHM**	KTR18EZPF3001
R405	300R	Resistor, SMD, 300R, 1%, 1/4 W	(1206)	**ROHM**	KTR18EZPF3000
R406	2.2K	Resistor, SMD, 2.2K,1%,1/8 W	(0805)	**ROHM**	KTR10EZPF2201
R407	75K	Resistor, SMD, 75K,1%,1/8 W	(0805)	**ROHM**	KTR10EZPF7502
R408	1.2K	Resistor, SMD,1.2K,1%,1/8 W	(0805)	**ROHM**	KTR10EZPF1201
R409	10K	Resistor, SMD,15K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF1002
R410	15K	Resistor, SMD,15K,1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF1502
R411	0R	Resistor, SMD, 0R,1%, 1/4 W	(1206)	**ROHM**	KTR18EZPJ000
R412, 413, R414, R415	499K	Resistor, SMD, 499K, 1%, 1 /4 W	(1206)	**ROHM**	KTR18EZPF4993

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R416	14.3K	Resistor, SMD, 14.3K, 1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF1432
R417	39K	Resistor, SMD, 39K, 1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF3902
R418	3.9K	Resistor, SMD, 3.9K, 1%, 1/8 W	(0805)	**ROHM**	KTR10EZPF3901
R419	100R	Resistor, SMD, 100R, 1%, 1/4 W	(1210)	**ROHM**	ESR25JZPF1000
RS400	1.5R	Resistor, SMD, 1.5R,,1%, 1 W	(2512)	KOA Speer	SR733ATTE1R50F
TR400	EE20, Horizontal	Transformer, TH, EE20, Horizontal, EE20	(10 Pin)	Custom	Custom
U400	BM2P063HK	DC/DC IC, SMD, IC, BM2P063HK, 650V, 40 ~ 105 °C	(DIP-7)	**ROHM**	BM2P063HK
U401	TL1431CL3T	Low Voltage Precision Adjustable Shunt Regulator, SMD, TL1431CL3T	(SOT-23-3)	STMicroelectronics	TL1431CL3T
U402	MCP1826S/3.3V	LDO, SMD, IC, MCP1826S/3.3V, 3.3V, 40C to +125C	(SOT-223-3)	Microchip	MCP1826S
Z400	18V	Diode, SMD, ZENER,18V	(SOD-323-2)	**ROHM**	UDZVTE-1718B

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9. Layouts & Assembly

Power Board

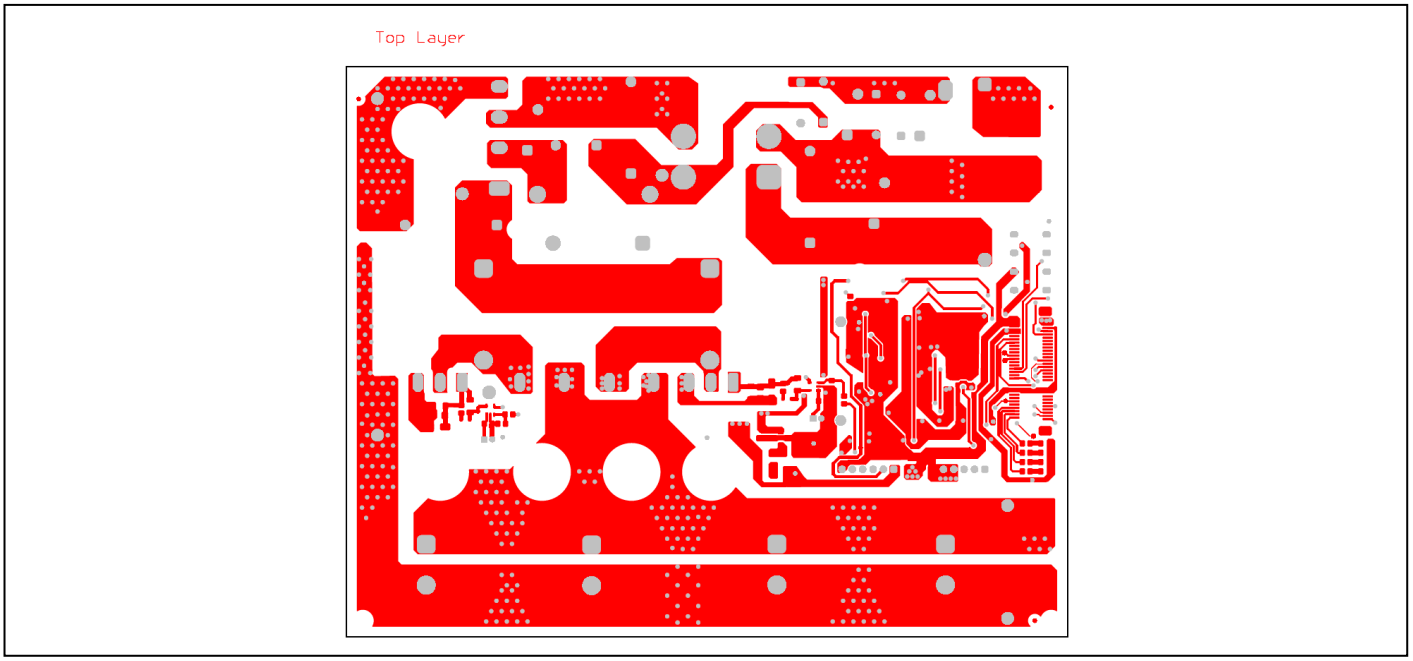


Figure 15. Top layer of the Power Board.

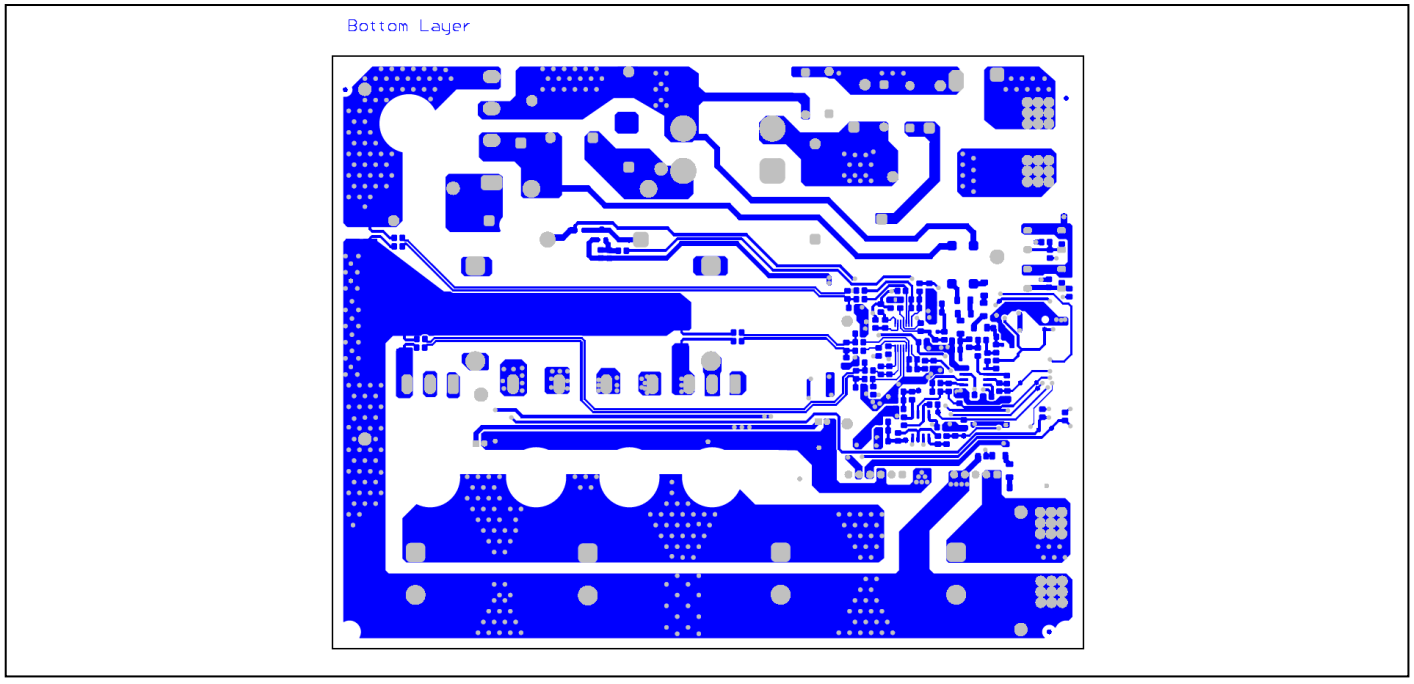


Figure 16. Bottom layer of the Power Board.

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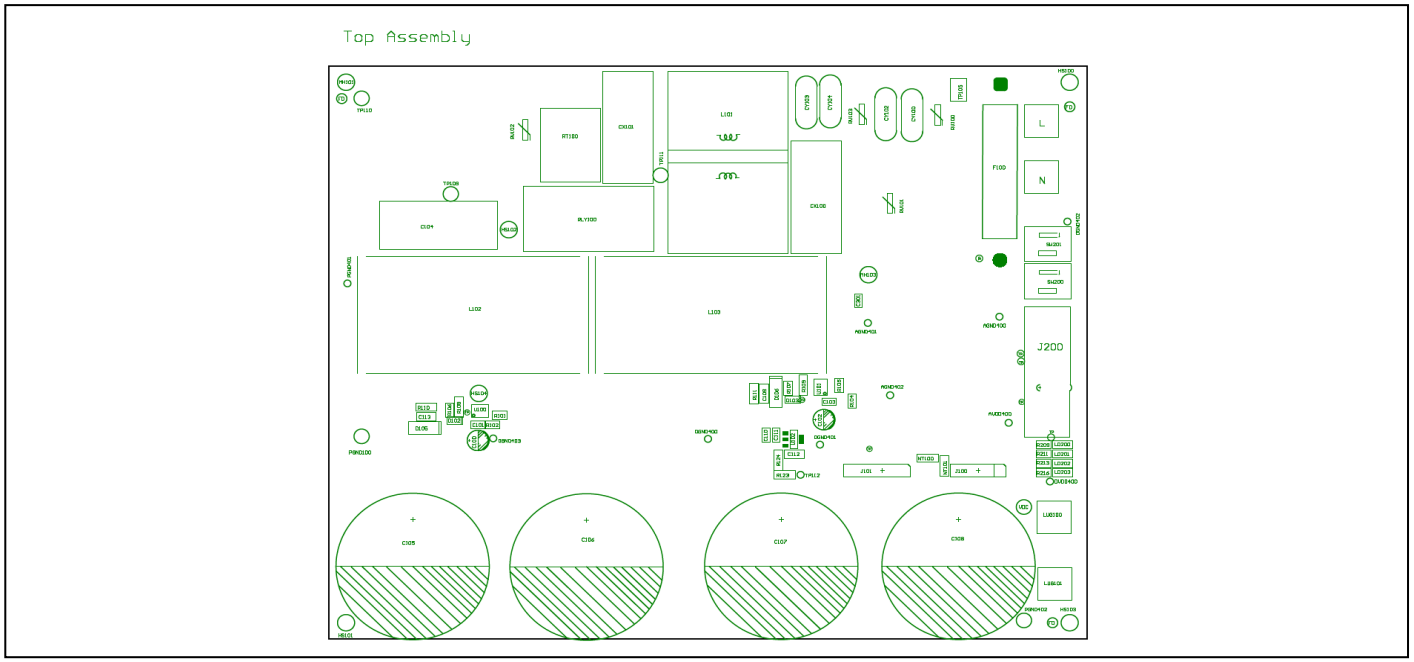


Figure 17. Top assembly of the Power Board.

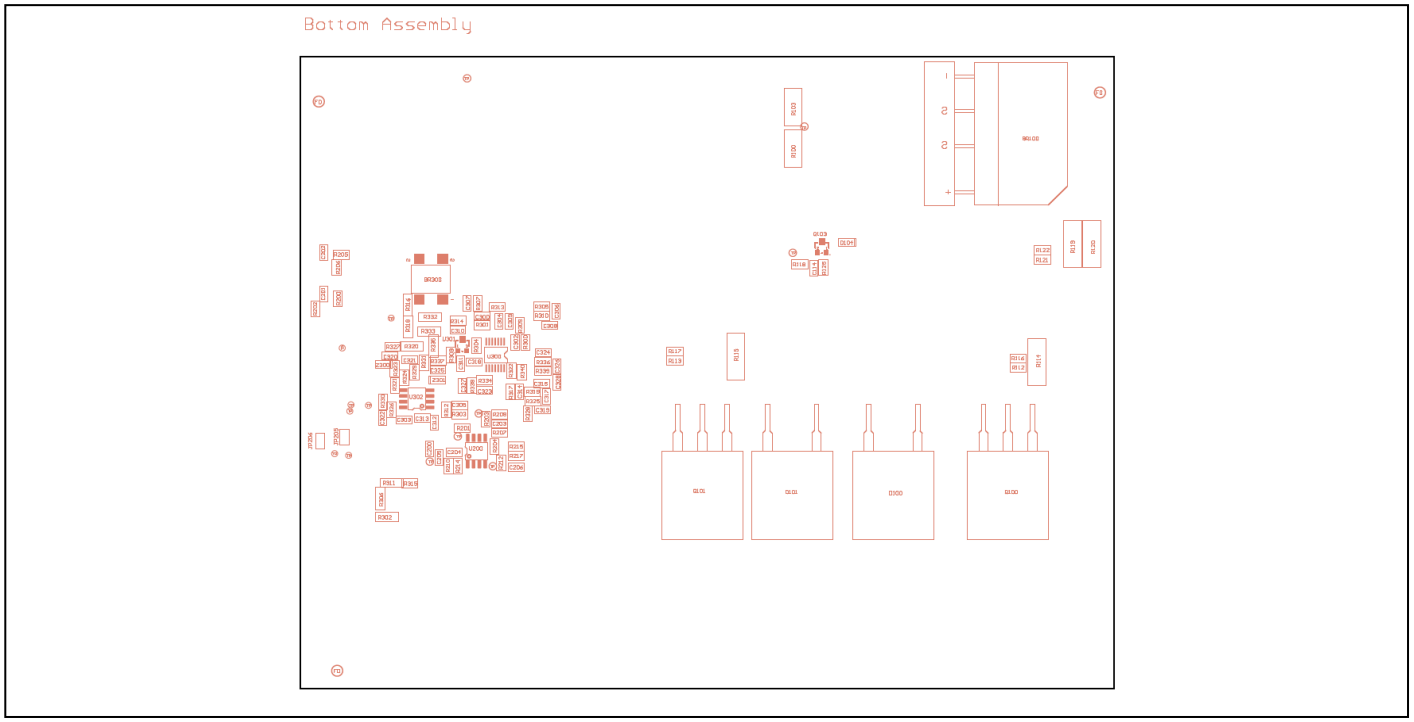


Figure 18. Bottom assembly of the Power Board.

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Auxiliary Power Supply (AuxPS)

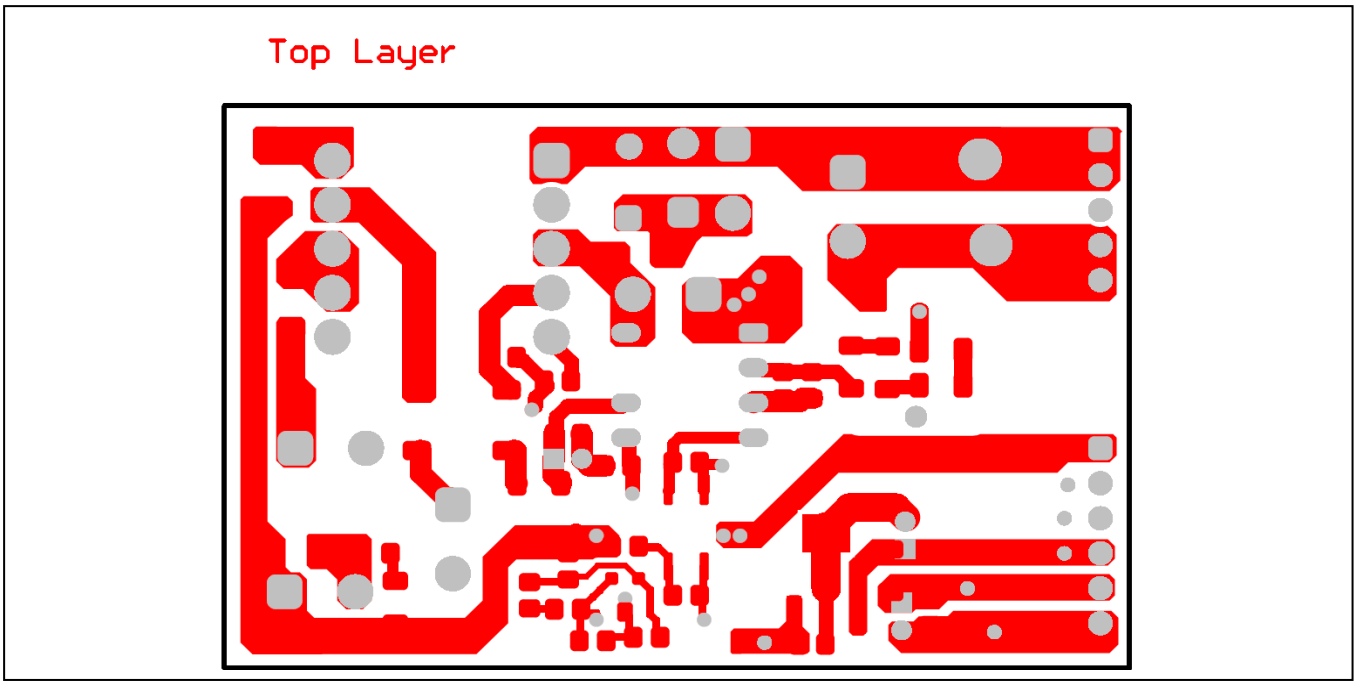


Figure 19. Top layer of the Auxiliary Power Supply (**AuxPS**).

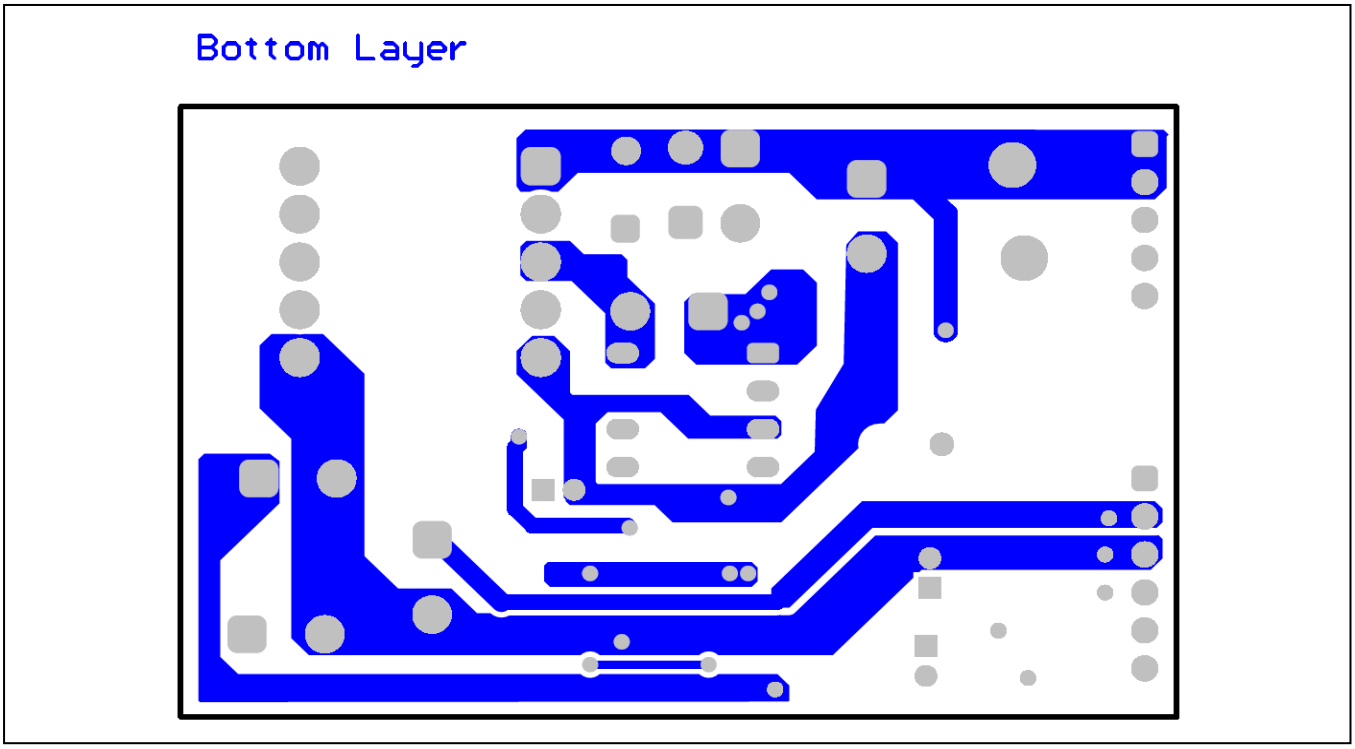


Figure 20. Bottom layer of the Auxiliary Power Supply (**AuxPS**).

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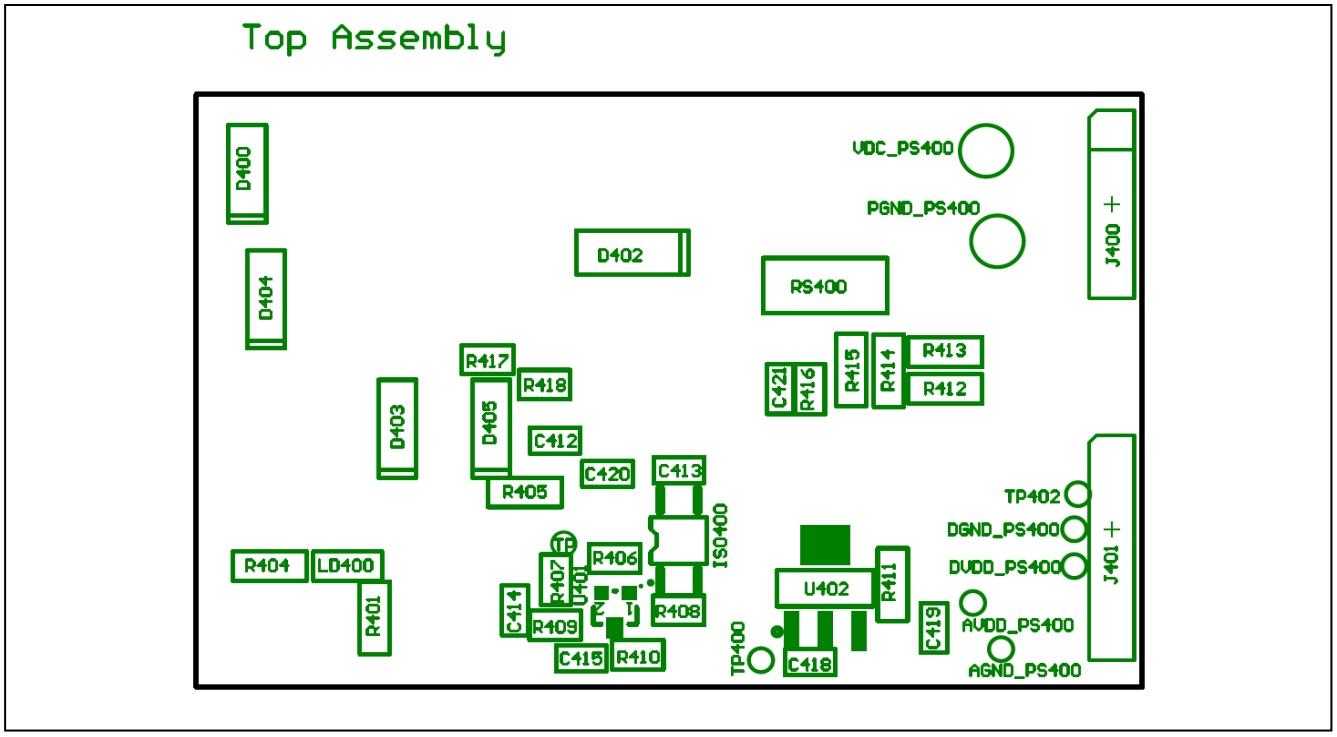


Figure 21. Top assembly of the Auxiliary Power Supply (**AuxPS**).

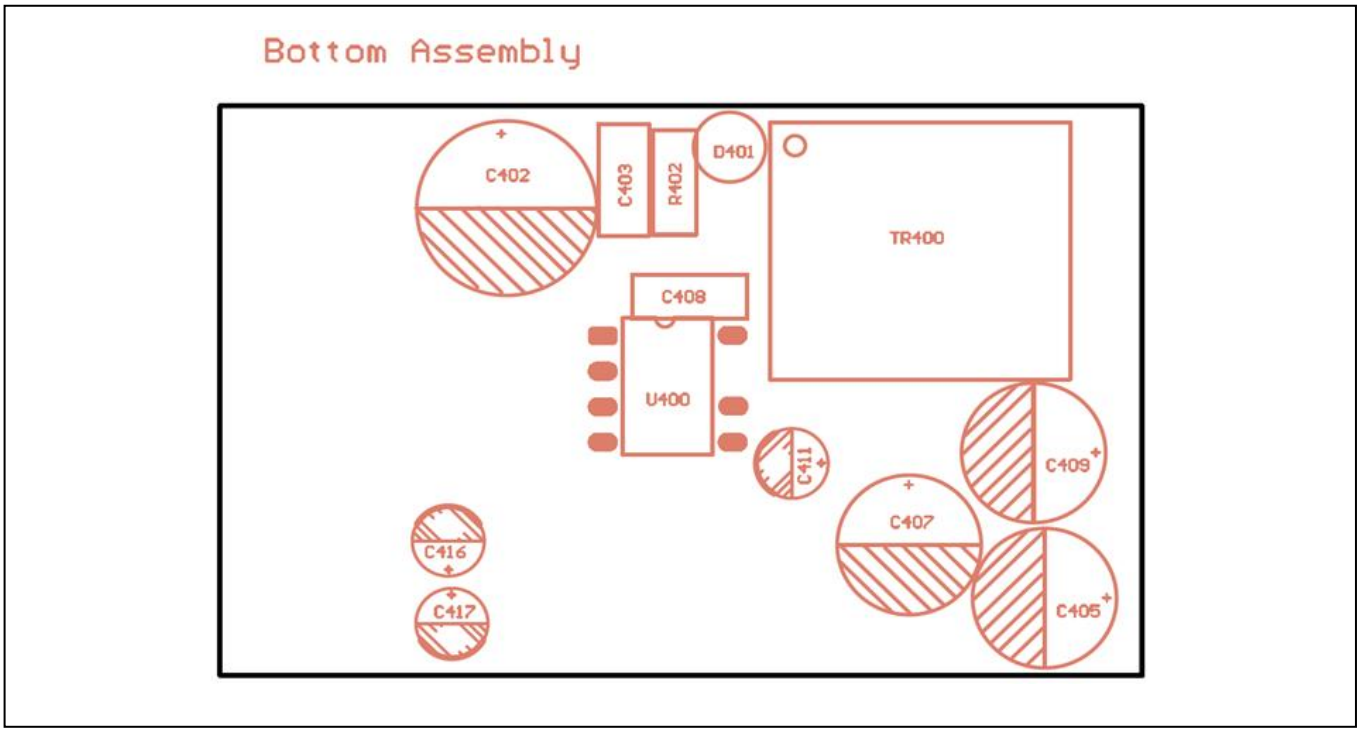


Figure 22. Bottom assembly of the Auxiliary Power Supply **(AuxPS)**.

Control PCB

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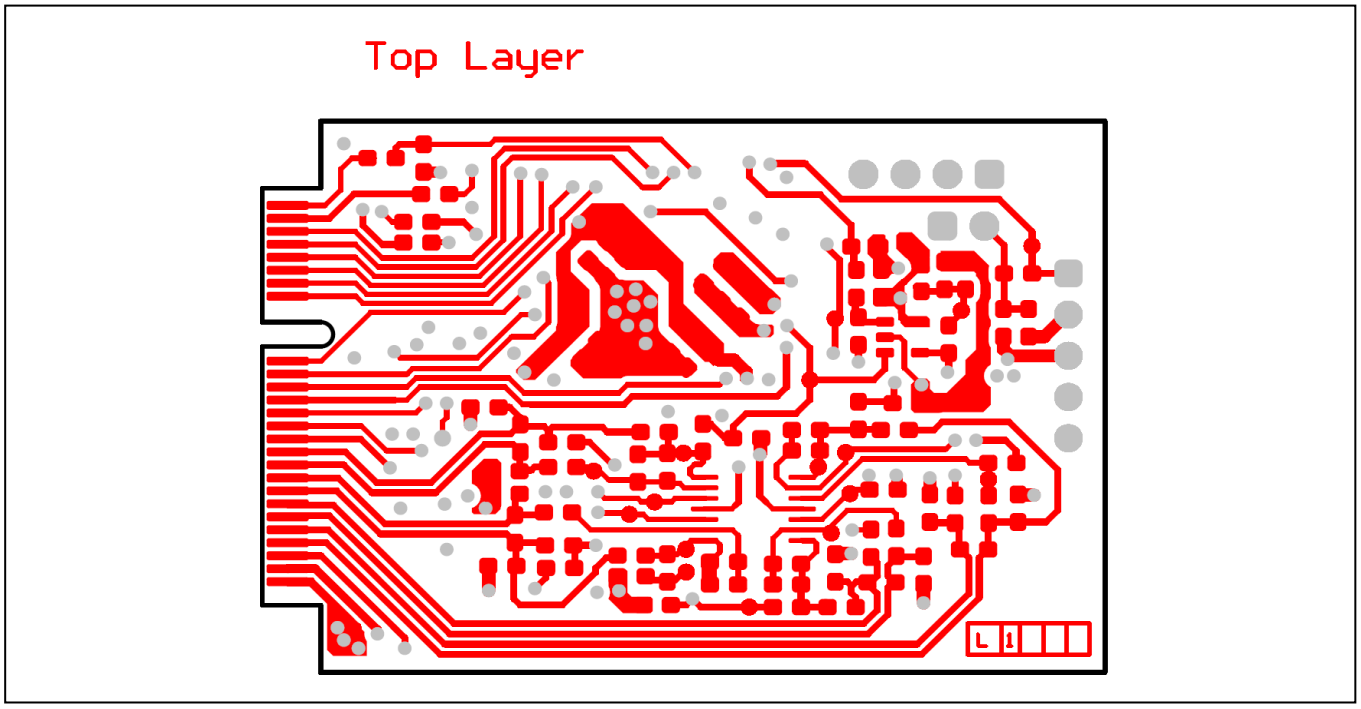


Figure 23. Top layer of the Control PCB.

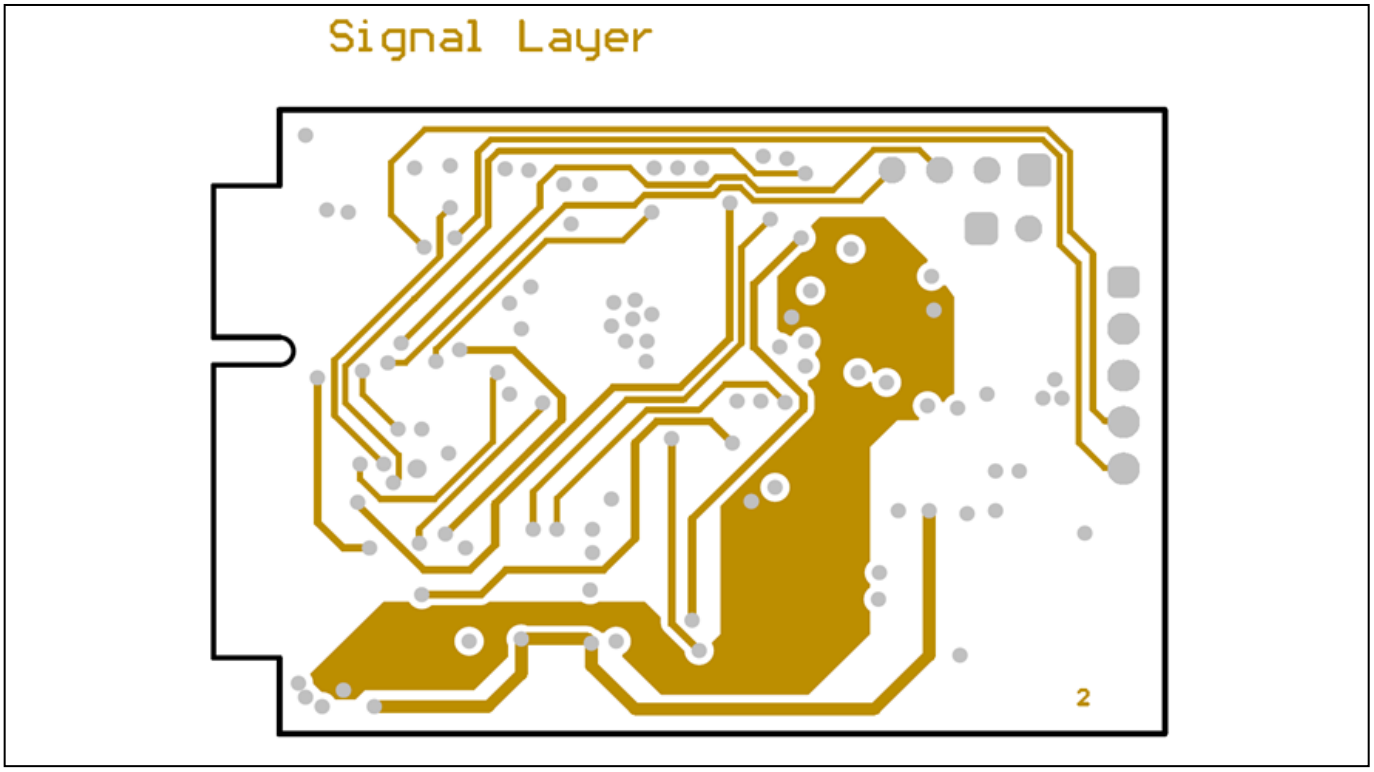


Figure 24. Signal layer of the Control PCB.

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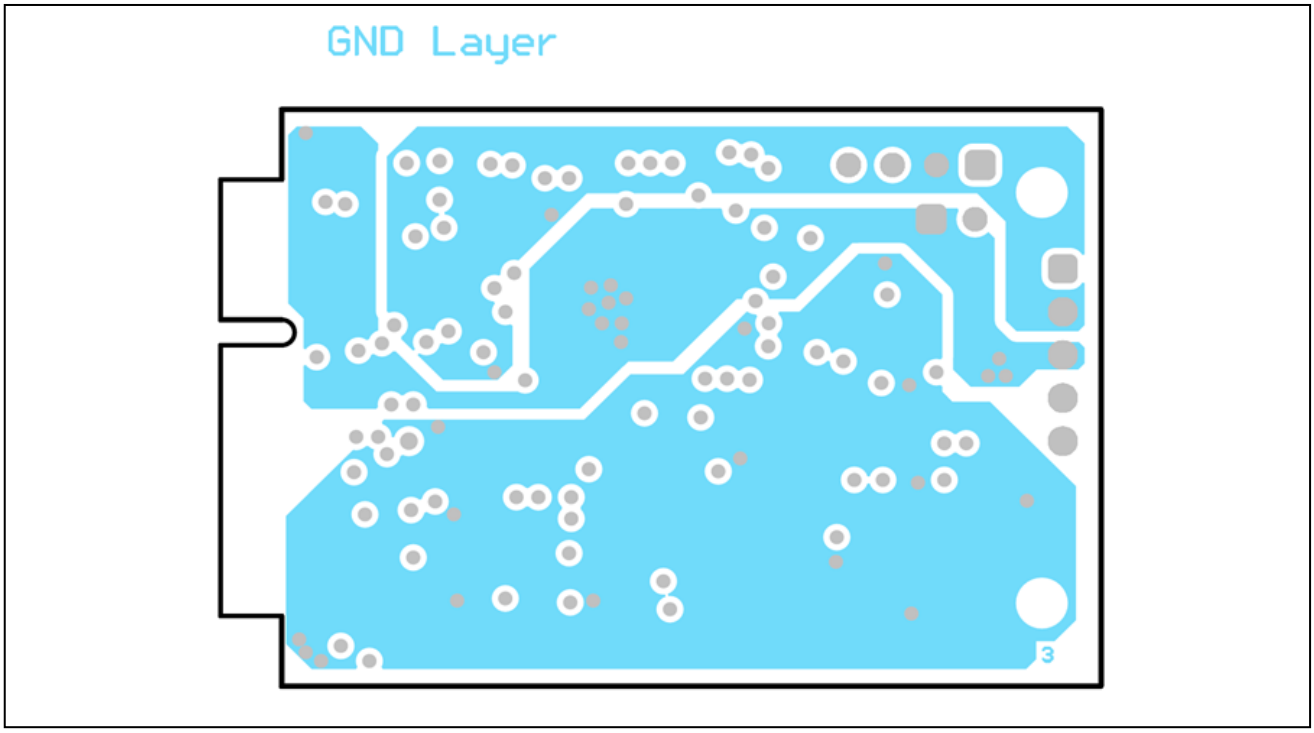


Figure 25. Ground layer of the Control PCB.

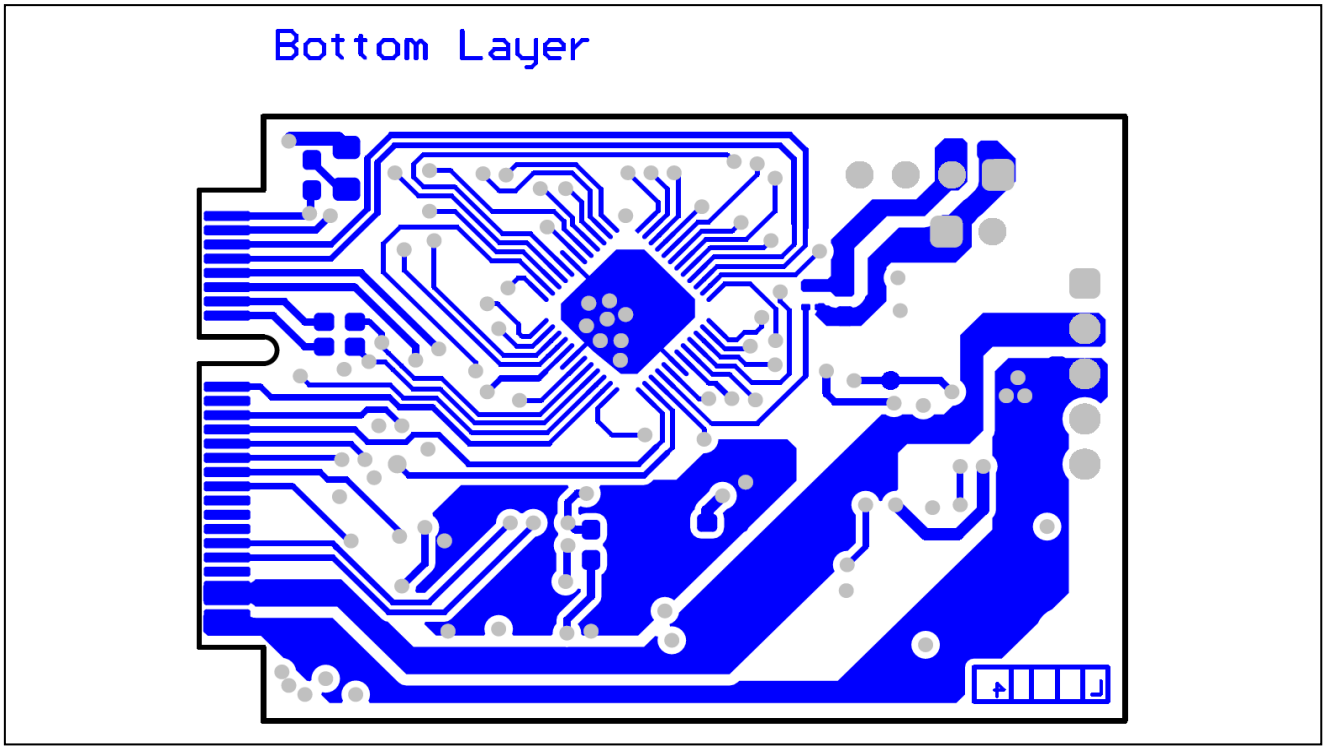


Figure 26. Bottom layer of the Control PCB.

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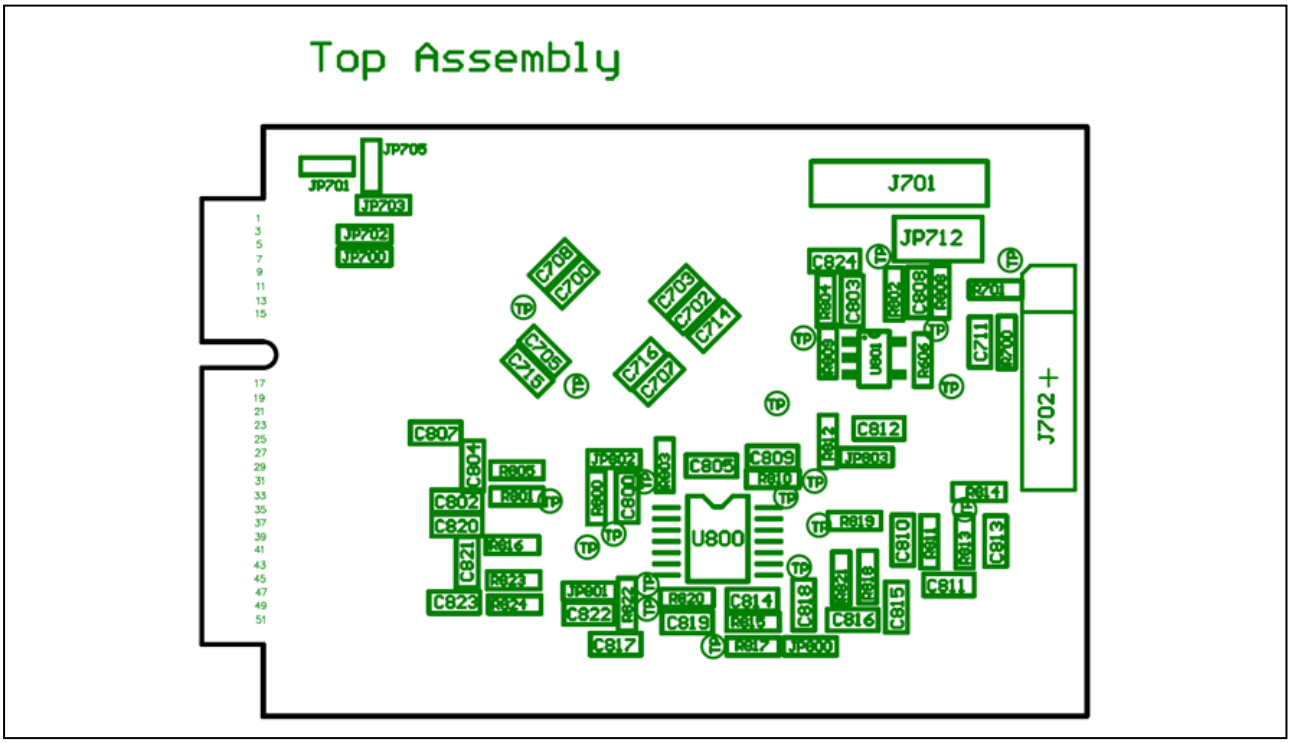


Figure 27. Top assembly of the Control PCB.

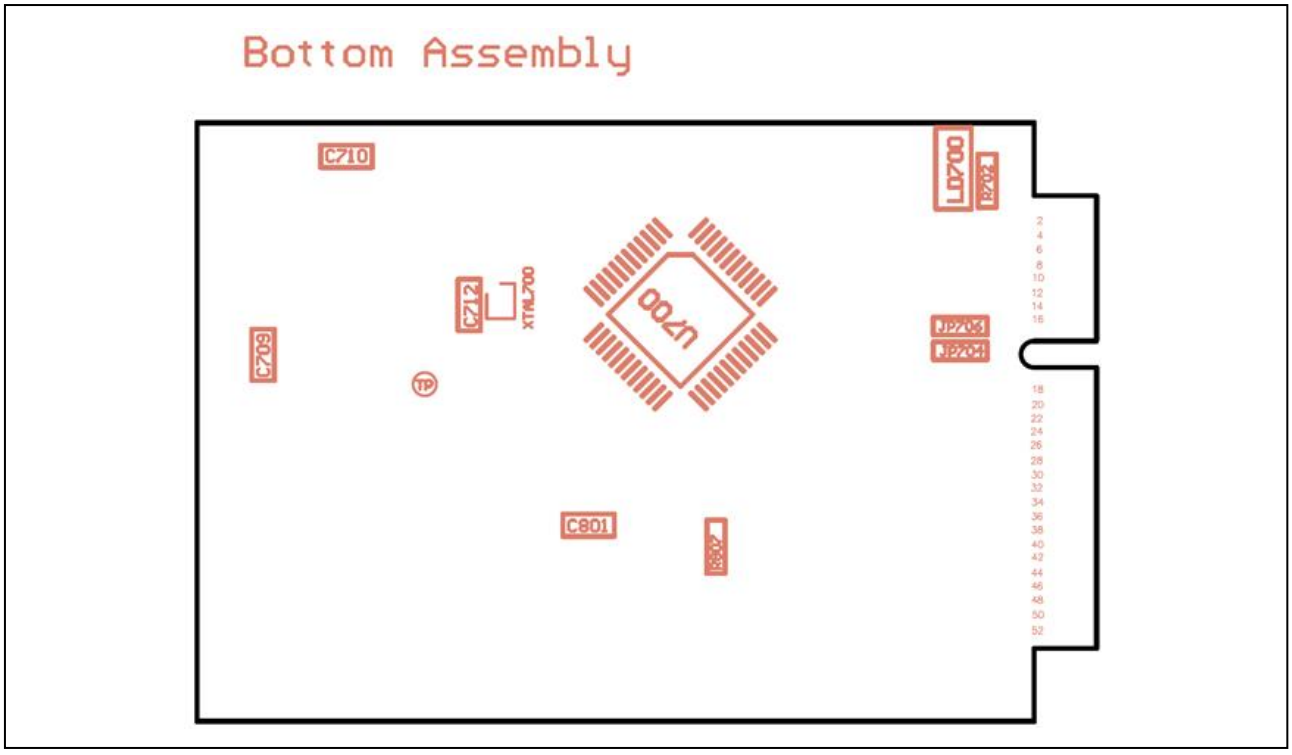


Figure 28. Bottom assembly of the Control PCB.

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